A New Experimental Method for Estimating Demand for Non-market Goods With an Application to the Value of a Statistical Life*

Grady Killeen

The University of Chicago

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Abstract

This paper introduces a new method for estimating demand for non-market goods and applies it to the value of a statistical life (VSL) in Kenya. The approach updates beliefs about the life-saving efficacy of a product (helmets) and elicits product choice. This allows demand to be estimated from subjective beliefs rather than assuming rational expectations. The resulting VSL estimate is $523, near the lower end of Kenyan estimates. Standard methods produce skewed results, driven by violations of rational expectations. The findings rationalize weak demand for preventive health and highlight limited evidence and theoretical support for VSLs used in benefit-cost analysis.

Keywords: Non-market amenities, value of a statistical life, rational expectations

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Email: gkilleen@uchicago.edu

1 Introduction

Many goods that are important to welfare — such as clean air and water, and health — are not directly traded. Pricing these non-market goods or amenities is important to design public policies, but the absence of markets makes it difficult to identify consumer preferences. This paper introduces a new experimental method to estimate demand for non-market goods by revealed preference and applies it to estimate consumers' willingness to pay to reduce mortality risk. My approach draws on the fact that even though these amenities are not traded, products often change exposure to or consumption of non-market goods (e.g. air purifiers).

Attempting to exogenously vary exposure to non-market goods is often infeasible. Rather, my approach shifts beliefs about the extent to which a given consumer product affects exposure to a non-market good. Demand for the consumer product then enables an inference about how the non-market good affects consumer utility. I estimate demand for the non-market good by examining how willingness to pay for the product changes with perceived exposure to the amenity, using the randomized provision of information to instrument for subjective beliefs. This method does not assume rational expectations, a common requirement of traditional approaches. It also does not require that agents fully update beliefs in response to information, or that individuals report their subjective beliefs without error. The primary identifying assumption is that measurement error in reported expectations (the gap between an agent's measured and true subjective belief) is classical.

I design two tests for bias created by non-classical measurement error in reported beliefs. One leverages multiple information treatments to test whether estimates scale with changes in beliefs, helping rule out experimenter demand effects. The second adapts insights from the local average treatment effects literature, reweighting observations as a function of priors to detect misreporting correlated with true beliefs, such as rounding or difficulty with small probabilities.

I apply this method to study demand for mortality risk reduction, the value of a statistical life (VSL), in Kenya. VSL is important for public policy but challenging to estimate. Mortality risk is a feature of economic decisions including healthcare, environmental regulation and transportation. Policymakers directly use VSL in benefit-cost analyses, and NGOs rely on the parameter to allocate development aid. In theory, VSL estimates allow practitioners to use consumers' own preferences for trading off mortality risk and consumption to guide policy, maximizing welfare and avoiding

paternalism. But economists have raised concerns that methods typically used to estimate the parameter may be prone to bias, causing misallocation (Ashenfelter, 2006).

VSL is often estimated by taking decisions where mortality risk is salient, then estimating demand models where the level of mortality risk present in data proxies for beliefs. This approach is biased if agents have beliefs about risk that differ from statistical estimates (i.e. their beliefs violate rational expectations). Moreover, estimates may be prone to omitted variable bias if mortality risk is correlated with other determinants of utility (Ashenfelter and Greenstone, 2004). Many studies also estimate VSL over highly selected populations where there is quasi-random variation in risk. Consistent with potential bias, estimates of VSL vary by orders of magnitude: in Africa, they range from below $1,000 to $700,000 (Kremer et al., 2011; León and Miguel, 2017).[1]

In this paper, I estimate the VSL of urban Kenyans by presenting motorcycle taxi passengers with randomly assigned information about the efficacy of helmets at preventing death. I then offer the choice between a helmet or cash. The approach does not rely on rational expectations, and the random assignment of information ensures that selection into risk does not bias estimates. Respondents were assigned to an information control or one of two treatments. One treatment arm was presented with estimates of unhelmeted mortality risk and the results of Liu et al. (2008), which estimates that helmets reduce one's risk of dying by 42%. The second received the same risk information, but a 70% helmet effectiveness estimate (Ouellet and Kasantikul, 2006), which differs because experts are unsure how well helmets work. I then elicited respondents' beliefs about the likelihood that a helmet would save their life and estimated helmet demand using a Becker et al. (BDM, 1964) mechanism, inverted to offer a free helmet or cash so that demand is not limited by cash on hand. I estimate VSL by fitting a regression of willingness to pay on the probability that a helmet will save the agent's life, using treatment assignment to instrument for beliefs.

I study the Kenyan motorcycle taxi market because helmet use is low, enabling VSL estimation over an informative population. Traffic accidents are a leading cause of death in Kenya and motorcycle taxi use is widespread. Yet only 3% of passengers wore helmets near the time of the study (Bachani et al., 2017). Since fatality risk rises with trip volume, ridership provides cross-sectional variation in mortality risk, which facilitates comparison with typical VSL estimation approaches.

[1] Stated preference VSL elicitation does not assume rational expectations, but agents have no incentive to answer truthfully and estimates may be prone to desirability bias, so I focus on revealed preference methods.

In the first main finding of the study, I estimate low demand for marginal reductions in mortality risk in this sample, with a VSL of 2022 PPP USD $523.[2] Receipt of the 42% helmet effectiveness estimate lowers average beliefs about the likelihood that a helmet will save one's life (a product of accident risk and helmet efficacy) from 2.7% to 1.9%, reducing demand for a helmet by about $3. My preferred estimate excludes values below $214 and above $1,043 with 95% confidence.

This VSL, while surprisingly low and smaller than values used in policy, is readily reconciled with values orders of magnitude larger among high-income populations. VSL is inversely proportional to the marginal utility of consumption. Reconciling this estimate with a USD 2022 $700,000 VSL of León and Miguel (2017), whose sample has incomes near $75,000, requires a coefficient of relative risk aversion of 3, within the range of standard macroeconomic estimates (Havránek, 2015).[3] Heterogeneity in this sample reflects this relationship, with VSL roughly $500 higher among respondents with above-median income. Prior evidence from Kenya supports this consumption utility explanation: Kremer et al. (2011) attribute a similarly low VSL (within the confidence sets I estimate) to a high income elasticity of demand for health.

The low VSL estimates also align with evidence of low demand for preventative health in LMICs. Cohen and Dupas (2010) find that only 40% of pregnant Kenyan women purchase a malaria net for $0.60, implying a cost per life saved under $200 (Pryce et al., 2018). Banerjee et al. (2010) likewise show that in-kind incentives worth $4 substantially increase childhood vaccination. Low demand could reflect decision-making errors or binding constraints. But adoption has often remained low when such frictions are targeted (Dupas and Miguel, 2017).

The second component of this study tests the identifying assumption that beliefs are measured without non-classical error. Two leading threats are experimenter demand effects and systematic misreporting. Both concern a gap between reported and decision-relevant subjective beliefs: the method allows beliefs to differ arbitrarily from the empirical truth. Two features of the design help test for bias. First, the treatments provide signals of varying intensity. Under classical error, VSL estimates should be unchanged if an arm is excluded. Otherwise, one would expect values to differ, since posteriors are centered at different points across treatments, unless misstatement is proportional across arms. Using any combination of arms to instrument for beliefs yields similar

[2] I use the World Bank's 2022 PPP conversion factor to convert to USD (42.93).

[3] I use this estimate for comparison because it is the highest-income revealed preference estimate obtained in an African transport setting. VSL estimates from the United States are higher still.

estimates. Helmet demand also falls in the low treatment and rises with elicited beliefs in the cross-section, providing revealed preference evidence that elicited beliefs are decision-relevant.

Another design feature that allows for tests of misreported beliefs draws on the logic of local average treatment effects. Instrumenting for posteriors using treatment assignment weights observations linearly in priors, but one may construct instruments that weight observations proportionally to priors squared, yielding similar results. The stability of estimates suggests findings are not substantially affected by any distortion that varies with the level of perceived risk, including rounding and difficulty reporting small probabilities.

Three further checks address threats to identification, and none changes the conclusion that VSL is low. First, I test whether difficulty reporting small probabilities biases estimates by eliciting perceived risk as the number of deaths per 10,000 riders, permitting integer responses. Second, I test for demand effects by comparing helmet demand in the control arm with a pure control not asked about mortality risk. Third, I assess bias in willingness to pay measures by estimating VSL with an IV probit that uses realized helmet receipt rather than elicited valuations.

The third part of the paper tests if rational expectations, an assumption in most VSL estimates, holds in this setting. I find systematic departures, with agents learning from their own experience and that of their social network rather than from representative data. Respondents who know someone involved in a motorcycle accident report substantially higher five-year risk. But the level of reported risk is essentially uncorrelated with the empirical risk implied by ridership. Respondents also report twice the risk when surveyed on a rainy day, which should have no bearing on risk over five years. Such departures are likely to undermine standard estimators. Consistent with such bias, two common approaches produce estimates ranging from -$6,167 to over $381,000 in this sample.

These results indicate that benefit-cost analyses conducted in developing countries may poorly reflect preferences, causing misallocation. Governments face the problem of funding health programs by raising taxes. Selecting programs that maximize welfare requires knowledge of how consumers trade off current consumption against uncertain future safety benefits, which VSL captures. Estimates of VSL are often low in poor countries (Jack and Ryan, 2026). But in practice, policymakers adjust rich country estimates by a fixed income elasticity to value lives saved in LMICs, resulting in higher VSLs. Applied in place of these values, my estimates move the benefit-cost ratios of four of five recent Kenyan analyses from above one to below one.

This paper's estimates — applied within the prevailing benefit-cost analysis paradigm — therefore suggest that redirecting funding from health programs towards poverty alleviation could increase welfare. An alternative view is that the finding that VSL is so low is symptomatic of underlying issues with VSL or its application. I discuss two possibilities. First, mortality risk may account for a smaller share of the total benefit of health programs in LMICs, in which case indexing the benefits of health programs to VSL would understate their returns. Second, practitioners may be applying the parameter outside the scope of behaviors about which VSL is informative.

VSL captures how much consumption someone is willing to forgo for a small reduction in their mortality risk. This is appropriate for valuing lives saved by locally financed programs, and for allocating aid between health and consumption programs within a setting. But when health investments are donor financed, recipients often face no consumption tradeoff. Applying recipient VSL in these cases will underestimate welfare gains by a factor of the ratio of recipient to donor marginal utility of consumption. In practice, recipient VSL is nonetheless standard. The OECD, World Bank, and US EPA follow this practice, and nearly all applications of VSL to externally funded programs in NBER working papers published since 2000 use recipient VSL. Because the VSL I estimate is so low, this practice can make programs that raise welfare fail a benefit-cost test. I propose the use of donor VSLs in such cases, and show that under standard estimates of the curvature of consumption utility this reduces worst-case welfare losses relative to current practice.

This paper is among the first to estimate VSL using subjective beliefs and with methods that are robust to omitted variable bias.[4] This addresses two leading concerns with traditional methods. The results also provide one of the first tests of rational expectations in the VSL literature. This builds on Ashenfelter and Greenstone (2004) and Greenberg et al. (2021) which study endogeneity and heterogeneity in VSL. It also relates to Baylis et al. (2023) which study imperfect information in the setting of air quality. The VSL estimates are also independently important because knowledge of demand for safety in LMICs is limited (Greenstone and Jack, 2015). I produce one of the first urban estimates using a design that ensures results are not determined by cash on hand.

This method may also be useful for valuing other non-market amenities. Goods such as environmental quality are economically important but difficult to value (Campbell and Brown, 2003; Greenstone and Jack, 2015). Policy decisions often rely on non-incentivized estimates (e.g. contin-

[4]Shrestha (2020) attempts to estimate VSL using similar methods, but confidence sets are uninformative.

gent valuation surveys) because of insufficient variation for revealed preference measures (Mendelsohn and Olmstead, 2009). Methods also tend to assume rational expectations, contradicting recent evidence (Baylis et al., 2023). The method I introduce could be applied to value such goods, provided that products exist that have known impacts on the amenities but have not been widely adopted. Section 6.3 details these conditions, and highlights products that could be used to study demand for clean air, clean water, or reduced morbidity in LMICs, where product diffusion is slow.

2 Study design and context

2.1 Motorcycle taxis and helmet use in Kenya

This study considers a sample of motorcycle taxi passengers in Nairobi, Kenya. This setting has three features well-suited to study demand for mortality risk reduction. First, motorcycle taxis are widely used and passengers rarely wear helmets, so VSL may be estimated over an informative sample. Second, helmets substantially reduce one's risk of death, allowing one to estimate demand for a substantive safety improvement. Third, empirical mortality risk varies across the sample since it is a function of ridership. This facilitates comparisons to typical methods for estimating VSL.

Motorcycle taxi use is common in Kenya. There are an estimated 2.4 million drivers providing taxi services, combining for 22 million trips/day.[5] Traffic accidents are a top-five cause of death.[6] Despite high risks, helmet use among taxi passengers is rare, measured at just 3% (Bachani et al., 2017). Low helmet use suggests that demand may be low. However, safe helmets only recently became affordable. The *FIA Foundation* began offering low-cost helmets in 2021 (a year before the experiment), near the same time a local producer entered the market.

A concern with this sample is that consumers with low VSLs may select into motorcycle use. However, most Kenyans use motorcycle taxis, limiting extensive margin selection. I am not aware of representative data capturing the share of consumers that use motorcycle taxis (transportation surveys often focus on primary commuting mode). But the Kenyan field team estimated that at least 85% of urban Kenyan adults use motorcycle taxis. Back of the envelope calculations also suggest that ridership needs to be high among urban adults to rationalize aggregate usage.[7]

[5] Fred Matiang'i, "The urgency of bodaboda reforms", *Nation.Africa*, 2022.

[6] "New initiative to tackle road crash deaths in Kenya," *World Health Organization*.

[7] There were 22 million trips/day in 2022, and a population of 53 million. 40% of Kenyans are farmers and at least 10% are under 5. Given 7.5 trips/week, this suggests 80% of urban adults use motorcycle taxis ($22 \approx 53 \cdot 0.5 \cdot \frac{7.5}{7} \cdot 0.8$).

Intensive margin ridership may be more selected. For instance, wealthier users may take more trips, and there is some evidence of this in the data. A strength of this setting is that ridership is observed, allowing one to re-weight estimates to limit any effect of ridership on sample composition. Section 5.1 shows that doing so has no statistically significant effect on results, empirically suggesting that ridership selection does not meaningfully affect VSL estimates in this context.

2.2 Recruitment

This study recruited consumers from motorcycle taxi stands in Nairobi. Surveyors censused 188 stands and conducted surveys at 97, selected for geographic coverage. Areas with high crime, which are poorer than average, were excluded for safety. Wealthy areas were not excluded, but rich neighborhoods often lack taxi stands or public roads, so are naturally under-represented. Survey locations are plotted in Appendix Figure A1.

The study used passenger arrival times to sample from the population at each location. Surveyors attempted to recruit the first individual to arrive at a stand after completing a survey. Those that did not report regular access to a helmet were informed that they could choose a free helmet or a cash gift if they completed a 15-30 minute survey. The high value of the gifts (about $15 on average) yielded a high response rate. Over 90% of passengers agreed to take part in the survey.[8]

Demographic information is presented in Table 1. The data suggests that the study reached a broad, albeit not representative, population. Income is close to county-level figures for Nairobi: the mean annual income is USD PPP $7,422, compared with gross per capita production of $7,907 for Nairobi county in 2017 reported by the Kenyan National Bureau of Statistics.[9] More men (981) than women (444) were surveyed, likely because men spend more time away from home in urban Kenya. Education is relatively high, with an average of about 12 years of schooling completed.

2.3 Information treatment: Motorcycle fatality risks and helmet effectiveness

This study implemented a randomized information treatment to produce variation in beliefs to estimate VSL. There are four arms: a pure control, control, and two treatments. The treatments were presented with information about the mortality risk of motorcycles and the life-saving effectiveness

[8]Surveyors reported recruitment rates and the share screened out due to helmet access to the field manager. The field manager reported that under 1% of approached passengers were screened out due to helmet access.

[9]Source: 2019 Gross County Product Report and 2017 World Bank PPP conversion rate.

of helmets, while the control arms received no such information.

The pure control was not asked about motorcycle safety. In contrast, the control was asked questions about the risks of motorcycles and their beliefs about the effectiveness of helmets. The pure control was included to test if asking respondents these questions affected demand. These observations are excluded from VSL estimates since beliefs were not elicited. In practice, and reassuringly, there are no differences in demand between the pure control and control.

The treatments presented different studies of helmet effectiveness. Respondents first received estimates of their unhelmeted mortality risk over the 5 year lifespan of a helmet.[10] Those in a "low treatment" arm were then given the results of Liu et al. (2008) which estimates that helmets reduce one's risk of dying by 42%. Respondents assigned to a "high treatment" arm were presented with an estimate from Ouellet and Kasantikul (2006) that helmets reduce fatality risk by about 70%. Surveyors followed standard scripts, so only the study and results vary across arms. The sources of information and the fact that the studies were conducted outside of Kenya were disclosed.

Neither arm was given information that I believed was inaccurate. There is a consensus that motorcycle helmets are effective, but there is uncertainty about exactly how well they work. The 42% figure presented in Liu et al. (2008) is often presented by the UN and NGOs operating in Kenya, motivating its use in this study. But their meta analysis primarily considers studies of high-income settings where driving speeds and helmet quality differ. Ouellet and Kasantikul (2006) present estimates from Thailand, which may be more similar in driving and helmet environment. It is unclear which estimate is more representative of the efficacy of helmets in Kenya.[11]

2.4 Helmet valuations

I measured helmet demand using an inverted Becker et al. (1964) elicitation. Respondents were asked the smallest cash transfer that they would prefer to a free helmet. Surveyors then revealed a random draw between 5 and 600 Kenyan shillings (Ksh). If the amount was at least the respondent's bid, they received cash. Otherwise they were given a helmet. Because respondents choose between two gifts, the elicited valuation corresponds to "chooser" valuations in Kahneman et al.

[10] I calculated per trip risk for the average Kenyan from NTSA data, then estimated 5 year risks based on the respondent's ridership. Mortality rates were obtained by dividing deaths in Kenya the year before the study by 22 million (trips/day) $\times$ 365. I combined the per trip estimate with 5 year ridership, obtained by scaling trips/week.

[11] An ethics appendix (Appendix E) details steps taken by the study to ensure that no respondents were harmed.

(1990). I refer to it as willingness to pay (WTP) for simplicity because such valuations closely track WTP rather than willingness to accept (WTA), which typically exceeds WTP due to endowment effects when agents give up an item they already own (Kahneman et al., 1990).

WTP elicitations can be biased by liquidity constraints and loss aversion over money, and by a mismatch between hypothetical and real choices (Murphy et al., 2005). The elicitation was designed to minimize these concerns. Offering a choice between cash and a helmet ensures that demand is not constrained by cash on hand and avoids loss aversion over either a helmet or cash the respondent already owns. Every choice was binding and the stakes were high, which avoids hypothetical bias, and evidence from related settings shows that such high stakes BDM valuations correspond closely to take-it-or-leave-it offers (Berry et al., 2020).

To the extent that this design differs from pure WTP, evidence suggests that it scales demand upwards (Kahneman et al., 1990), which would, if anything, overestimate VSL and leave the paper's main conclusions unchanged. Moreover, since all estimates in this paper use the same valuations, comparisons across methods would be unaffected. One caveat is that valuations of health goods that consumers already own may be higher due to endowment effects, which limits the applicability of these estimates to such goods. However, WTP is the policy-relevant measure for most VSL applications, which typically evaluate proposals for new policies.

The maximum draw was based on assessments by an NGO and the helmet manufacturer that most valuations would fall below Ksh 600. There were also concerns that draws above market prices may cause consumers to believe helmets were expensive, harming diffusion. The manufacturer sold the helmets used in this study at a price of Ksh 580, 15% of weekly median wages. A limitation of the BDM mechanism is that one's true valuation is not the only optimal bid if an agent *knows* their valuation exceeds the maximum draw. To hedge against this risk, the BDM script did not mention the upper bound or helmet value. In principle, one could infer the maximum draw from the consent form, which disclosed possible gifts but did not tie them to the game. The consent was presented well before the BDM exercise, making it unlikely that information would be recalled. In piloting, surveyors verified that respondents could not state the value.

In practice, helmet valuations often exceeded the wholesale price.[12] Appendix Figure A2

[12]High demand is reconciled with low VSL because consumers value helmets for reasons beyond life saving potential. Respondents frequently cited protection from non-fatal injuries, resulting in high hospital bills, during piloting.

demonstrates that there is no unusual behavior in bids near Ksh 600. In addition, agents in the low treatment arm were less likely to receive a helmet ($p < 0.1$, $p < .05$ if baseline risk is above 1 in 100,000), validating that valuations affect real outcomes. I also show that results are robust to using an IV probit estimator not influenced by variation in valuations above the maximum draw.

2.5 Randomization

Respondents were assigned to the information arms using a random number drawn in SurveyCTO. In wave one, respondents were assigned to the pure control with probability 0.1 and each of the other groups with probability 0.3. In wave two, the pure control was eliminated and respondents were assigned to the remaining arms with equal likelihood.

3 Model and identification

This section presents a simple model of demand for helmets and demonstrates how VSL is identified under the new experimental approach and typical methods.

Consider a population indexed by $i \in \{1, \ldots, \infty\}$. Agents maximize expected utility and Bayesian update their beliefs about motorcycle risks and helmet effectiveness. Learning is modeled to show how it can generate biased VSL estimates when beliefs are proxied by empirical risk. Identification requires only that information changes expectations, which I test empirically.

Each consumer has a prior about the probability of dying while wearing a helmet in a motorcycle accident that would be fatal without a helmet given by

$$Pr(D = 1|\mathcal{I}_i) \sim Beta(a_{iH}, b_{iH})$$

where $\mathcal{I}_i$ is one's information set and D is a Bernoulli random variable equal to 1 if the agent dies. Beliefs are over the unknown parameter of this distribution. There is true stochastic variation in whether agents in an accident die, but they must learn the parameter governing the process. The Beta distribution is natural because it is a conjugate prior of Bernoulli trials. a_{iH} may be interpreted as the expected number of deaths and b_{iH} the number of survivals out of $a_{iH}+b_{iH}$ accidents. Define

$$H_{i0} \equiv \mathbb{E}[Pr(D = 1|\mathcal{I}_i)] = \frac{a_{iH}}{a_{iH} + b_{iH}}$$

Suppose that the consumer receives a signal that the estimated efficacy of helmets is $\theta_H \sim Binomial\left(a_{EH}+b_{EH}, a_{EH}/(a_{EH}+b_{EH})\right)$. The Binomial likelihood captures the fact that studies of helmet efficacy report the number of deaths given a number of accidents. a_{EH} represents the number of fatalities and b_{EH} the number of survivals out of $a_{EH}+b_{EH}$ empirically recorded accidents. The agent's posteriors about the efficacy of helmets are

$$Pr(D=1|\mathcal{I}_i,\theta_H) \sim Beta(a_{iH}+a_{EH}, b_{iH}+b_{EH})$$

with expected value

$$H_{i1} \equiv \mathbb{E}[Pr(D=1|\mathcal{I}_i,\theta)] = \frac{a_{iH}+a_{EH}}{a_{iH}+a_{EH}+b_{iH}+b_{EH}}$$

If $\frac{a_{iH}}{a_{iH}+b_{iH}} \neq \frac{a_{EH}}{a_{EH}+b_{EH}}$, the empirical mortality rate differs from an agent's prior expectation, so the consumer's posterior mean will differ from their prior ($H_{i0} \neq H_{i1}$). If agents learn through own experience or their social network, beliefs will likely violate rational expectations. Learning from one's experiences is subject to survivor bias. Social network information is prone to small sample biases, and reports of fatal accidents may be more likely to spread.

The likelihood that a helmet saves one's life is a product of the probability of an accident and the effectiveness of a helmet given an accident. Therefore, suppose that the agent has a prior about the per trip risk of suffering an accident that would be fatal to an unhelmeted passenger given by

$$Pr(A=1|\mathcal{I}_i) \sim Beta(a_{iA}, b_{iA})$$

where A is a Bernoulli random variable equal to 1 if a trip ends in an accident. Beliefs are again over the unknown parameter of this Bernoulli distribution.

Suppose the consumer completes n_i motorcycle rides over the lifespan of a helmet. An unhelmeted agent's expectation of getting into a fatal accident over these trips is given by

$$r_{iu} = 1 - \mathbb{E}[(1-Pr(A=1|\mathcal{I}_i))^{n_i}] \tag{1}$$

with the expectation over the agent's beliefs. $(1-Pr(A=1|\mathcal{I}_i))^{n_i}$ is the probability of surviving

n_i trips given that the probability of suffering a fatal accident is $Pr(A = 1|\mathcal{I}_i)$ on any given trip. One must take the expectation over $Pr(A = 1|\mathcal{I}_i)$ since beliefs are not degenerate.

I assume that helmeted individuals involved in an accident that would otherwise be fatal stop using motorcycles.[13] Let z_i indicate whether the agent is exposed to the signal of helmet effectiveness θ. Their subjective probability of suffering a fatal motorcycle accident with a helmet is

$$r_{ih}(z_i) = \begin{cases} H_{i0} \cdot r_{iu}, & z_i = 0 \\ H_{i1} \cdot r_{iu}, & z_i = 1 \end{cases} \tag{2}$$

which is the unhelmeted risk times the perceived chance that a helmet fails to prevent death.[14] The subscript u refers to the agent's perceived unhelmeted risk and h to their helmeted risk.

Let p_i be the price of a helmet. The present value of utility from being alive is given by $u_a(x_i)$ where x_i is a vector of characteristics, such as income, health, and age. Denote flow utility of consumption by $u(c_i; x_i)$ and the expected utility from not being alive (which may be positive, e.g. if one believes in an afterlife) by $u_d(x_i)$. The agent's expected utility from purchasing a helmet is

$$U_{ih}(z_i) = \zeta_h + [1 - r_{ih}(z_i)] \cdot u_a(x_i) - p_i \cdot u'(c_i; x_i) + r_{ih}(z_i) \cdot u_d(x_i) + \epsilon_{ih} \tag{3}$$

ζ_h gives average utility from helmet attributes other than mortality risk, such as injury protection and comfort. ϵ_{ih} denotes idiosyncratic variation in utility. Without a helmet, expected utility is

$$U_{iu} = (1 - r_{iu}) \cdot u_a(x_i) + r_{iu} \cdot u_d(x_i) + \epsilon_{iu}$$

Setting $\epsilon_i = \epsilon_{ih} - \epsilon_{iu}$ and $\Delta r_i(z_i) = r_{iu} - r_{ih}(z_i)$,

$$U_{i,h-u}(z_i) \equiv U_{ih}(z_i) - U_{iu} = \zeta_h + \Delta r_i(z_i) \cdot [u_a(x_i) - u_d(x_i)] - p_i \cdot u'(c_i; x_i) + \epsilon_i$$

In words, one's expected utility from a helmet is a function of non-safety preferences for helmets, the agent's *belief* about the probability that the helmet will save their life, $\Delta r_i(z_i)$, times differences in expected utility from surviving versus not, less the price of a helmet times the marginal utility of consumption. The goal is to identify VSL, the change in income needed to compensate

[13] Otherwise, $r_{ih_0} = 1 - \mathbb{E}\left[(1 - H_{i0}Pr(A = 1|\mathcal{I}_i))^{n_i}\right]$, which is similar since suffering 2 accidents is unlikely.

[14] Agents may increase ridership if they perceive a helmet is more effective. Empirically accounting for a ridership response has little effect on estimates, so I omit this from the model for simplicity.

an agent for a change in mortality risk. Totally differentiating,

$$dU_{i,h-u} = \frac{\partial U_{i,h-u}}{\partial \Delta r_i} d\Delta r_i + \frac{\partial U_{i,h-u}}{\partial p_i} dp_i$$

Setting $dU_{i,h-u} = 0$,

$$VSL_i \equiv \frac{dp_i}{d\Delta r_i} = -\left(\frac{\partial U_{i,h-u}}{\partial \Delta r_i}\right) \Big/ \left(\frac{\partial U_{i,h-u}}{\partial p_i}\right) = \frac{u_a(x_i) - u_d(x_i)}{u'(c_i; x_i)} \tag{4}$$

This predicts that VSL will increase with income. Theory suggests that $u_a(x_i)$ will increase and $u'(c_i; x_i)$ will fall as agents become richer. $u'(c_i; x_i)$ is likely to be especially important as estimates indicate it falls non-linearly with income (Havránek, 2015). Expected utility from not being alive, $u_d(x_i)$, may also vary across contexts, especially with beliefs about an afterlife.

3.1 Identification under rational expectations

For simplicity, suppose that $\beta = u_a(x_i) - u_d(x_i)$ and $\alpha = u'(c_i; x_i)$ are homogeneous across agents. Let Δr_i^* be a statistical estimate of the likelihood a helmet will save one's life. A common set of identifying assumptions in the VSL literature (but not this paper) is

$$\text{Full Information Rational Expectations:} \quad \Delta r_i = \Delta r_i^* \quad \forall i \tag{5a}$$

$$\text{Exogeneity and Logit Errors:} \quad \epsilon_{ih}, \epsilon_{iu} \sim_{iid} EV1 \tag{5b}$$

Given data denoting if agents purchased helmets y_i, empirical risks Δr_i^* and prices p_i,

$$Pr(y_i = 1) = \Lambda\left(\zeta_h + \beta \cdot \Delta r_i^* - \alpha \cdot p_i\right) \tag{6}$$

Λ is the Logistic CDF and so α, β and thus VSL, which is $\frac{\beta}{\alpha}$, are identified.[15]

Assumption 5a fails if $Pr(\Delta r_i \neq \Delta r_i^*) > 0$, meaning agents hold beliefs about risk that diverge from statistical estimates, even if they match in expectation (in which case attenuation bias would affect estimates). This is plausible if consumers have the same information as researchers, but fails if they possess private information or lack access to an econometrician's data. Rational expectations has been assumed despite these limitations because of a lack of an incentive-compatible

[15] In practice studies often estimate mixed logit models, but the assumptions are similar.

alternative (prior to this paper) capable of producing informative estimates.[16] Assumption 5b is violated by omitted variable bias. For example, frequent motorcycle riders face higher risk but may also experience greater disutility from helmet discomfort.[17]

3.2 Identification using subjective belief data

This study estimates VSL without assuming rational expectations or exogenous mortality risk by using an instrument that shifts agents' subjective beliefs. The model shows that this is an appropriate way to identify VSL since presenting agents with information about helmet efficacy will change the expected utility they obtain from the life saving potential of a helmet (unless their priors exactly align with the signal or are degenerate). Updating beliefs about helmet efficacy produces the same identifying variation that exogenous differences in mortality risk would under rational expectations, but this approach is robust to biased beliefs and endogeneity. Formally, assume

$$\text{Exclusion Restriction:} \quad Cov\left(z_i, \epsilon_i\right) = 0 \tag{7a}$$

$$\text{Instrument Relevance:} \quad Cov(\Delta r_i, z_i) \neq 0 \tag{7b}$$

Letting v_i denote willingness to pay for a helmet, an agent's indifference between paying v_i for a helmet versus not gives

$$\begin{aligned} \epsilon_{iu} &= \zeta_h + \beta \cdot \Delta r_i(z_i) - \alpha \cdot v_i + \epsilon_{ih} \\ \alpha v_i &= \zeta_h + \beta \cdot \Delta r_i(z_i) + \epsilon_i \\ v_i &= \frac{1}{\alpha}\zeta_h + \frac{\beta}{\alpha} \cdot \Delta r_i(z_i) + \frac{1}{\alpha}\epsilon_i = \zeta_h' + VSL \cdot \Delta r_i(z_i) + \epsilon_i' \end{aligned} \tag{8}$$

VSL is identified from data on v_i and (potentially misreported) beliefs $\Delta r_i(z_i)$, using information exposure z_i to instrument for beliefs (ζ_h' is an intercept). When z_i is randomized, then denoting measurement error free beliefs by Δr_i^t, assumption 7a is equivalent to

$$\text{Classical Measurement Error:} \quad \mathbb{E}[\Delta r_i - \Delta r_i^t | z_i] = 0 \tag{7a'}$$

[16] For instance, León and Miguel (2017) "utilize a standard expected utility individual choice framework, using accident risk from historical data, in part due to the absence of a well-articulated and widely accepted alternative analytical approach that incorporates these behavioral concerns and generates meaningful valuation estimates."

[17] Some studies use instruments to relax 5b but still require rational expectations (e.g. Ito and Zhang, 2020).

This holds since z_i only contains information about Δr_i and (by randomization) is independent of other drivers of v_i. Therefore exclusion typically holds if misreporting is uncorrelated with z_i. Relevance holds as long as information changes beliefs (even if agents are not Bayesian). This may be empirically verified by testing if $Cov(\Delta r_i, z_i) \neq 0$.

The linearity of VSL in mortality risk follows from considering marginal reductions in risk. Applying VSL to large changes in mortality risk would likely underestimate welfare gains.

This framework may be used to estimate demand for other non-market goods by letting r_i be the probability of exposure to the good and z_i be an instrument shifting these beliefs. For instance, r_i may be the probability of exposure to unclean air mitigated by an air purifier.

4 Data and empirical specification

4.1 Data

I use data from 1,571 surveys collected in two waves. The first wave was conducted from October to December 2022, and the second between February and March 2023. The first wave included 921 surveys, counting pure control observations. The second included 650.

The survey collected detailed demographic data and information about the motorcycle taxi use of all respondents, and baseline and posterior beliefs about motorcycle taxi risks from those that were not assigned to the pure control group.

The methodology used to elicit beliefs was refined in piloting. Details of the final approach are presented in Appendix A and the full instrument is available on the AEA RCT Registry.[18] The survey elicited priors about per trip mortality risk, the average number of deaths per 10,000 motorcycle taxi passengers over 1 and 5 year periods, and the respondent's risk of dying in a motorcycle accident over 5 years. Posteriors about the respondent's 5 year mortality risk without a helmet and the effectiveness of helmets at preventing death were collected. Distinct measures were obtained to validate the variables against each other.

Briefly, priors and posteriors about unhelmeted risk were elicited by first having respondents select what range of probabilities their beliefs fell into from bands that span the unit interval (such as less than 1:10 million, 1:10 million to 1:1 million, up to greater than 1:10). Respondents were then asked to list a more precise belief in the interval. This approach reduced rounding in piloting.[19]

[18] The instrument is appended to the end of the PAP amendment.

[19] There is likely still rounding, especially of high beliefs. Such rounding likely results in classical error, but I test

Belief elicitation was not incentivized because the object of interest is respondents' perceptions of their own mortality risk. Incentivizing accuracy would require benchmarking against some average risk, inducing misreporting if respondents believed their own risk differed from the mean or mistrusted the data (e.g. anecdotally many respondents had concerns politicians under-count deaths). For beliefs about helmet effectiveness, the natural benchmark would be the published estimates presented in the treatments, so incentivizing accuracy would reward respondents for repeating the signal rather than reporting their belief, generating a first stage contaminated by experimenter demand effects. Identification requires that reports reflect perceptions up to classical error, and the study is designed both to validate that elicited beliefs affect real outcomes and to test whether misreporting affects VSL estimates. First, I demonstrate alignment between the multiple approaches used to elicit beliefs (Section 5.1). Second, helmet demand is increasing in beliefs about motorcycle risk and helmet effectiveness in the cross-section (Appendix Table A2). Third, demand falls in the arm that receives an efficacy signal below most respondents' priors (Appendix Table A7). Fourth, I show in Section 5.1 that exposure to an accident victim, one of the main sources of information respondents report using to form perceptions, moves both perceived risk and helmet demand by a statistically significant margin. Fifth, the study design includes multiple features to test for demand effects, poor comprehension of small probabilities, or other misreporting that could bias VSL estimates, as detailed in Section 5.2.

Table 1 summarizes sample demographics and balance across treatments. Appendix Table A1 similarly shows motorcycle use and beliefs. Variables are generally balanced across arms, although there is some imbalance among the pure control (which had a smaller number of respondents).

4.2 Experimental VSL estimation

The primary estimate of VSL is obtained via the two-stage least squares model

$$\begin{aligned} v_i &= \zeta_h' + VSL\Delta r_i + X_i'\gamma_0 + \gamma_1 r_{0,i} + \epsilon_i' \\ \Delta r_i &= Z_i'\pi + X_i'\pi_c + \pi_r r_{0,i} + \nu_i \end{aligned} \tag{10}$$

where v_i is the respondent's willingness to pay for a helmet, Δr_i is their perceived reduction in mortality risk from a helmet, X_i denotes controls, and $r_{0,i}$ is a measure of the respondent's baseline

for bias from non-classical error. Results are robust to including fixed effects for the interval selected.

belief about their unhelmeted mortality risk. I report results with two sets of instruments, Z_i. First, I consider an "interacted" set $Z_i = (T_i', r_{0,i} \cdot T_i')'$ where T_i is a vector of treatment assignment indicators. The second set consists of "treatment only," $Z_i = T_i$.

The preferred estimate uses the interacted instruments because they exploit heterogeneity in priors, improving power.[20] Since the perceived chance a helmet saves one's life is the product of perceived accident risk and perceived efficacy, the treatment shifts Δr_i by an amount proportional to $r_{0,i}$. T_i alone predicts the average shift; the interaction predicts each respondent's. Abadie et al. (2023) show that instruments of this form improve asymptotic mean squared error. Because $r_{0,i}$ is potentially endogenous, I control for it linearly, leaving only treatment-induced variation in $r_{0,i} \cdot T_i$. One concern is that if $r_{0,i}$ and Δr_i were elicited the same way, common measurement error would pass into the first-stage and attenuate estimates. I therefore construct $r_{0,i}$ from a distinct elicitation: the respondent's belief about the number of deaths per 10,000 motorcycle users over five years.

Controls are selected using single-post LASSO. Possible controls include demographic variables, motorcycle trip characteristics, and the information sources used to construct beliefs about mortality risk. Estimates also include enumerator fixed effects.[21] The primary tables report homoskedastic standard errors and weak instrument robust confidence sets. I also report heteroskedastic robust weak IV confidence sets for the primary VSL estimates.[22]

4.3 Alternative VSL estimation procedures for comparison

I compare the experimental VSL estimates to two existing approaches. First, I estimate

$$v_i = \zeta_h + VSL\Delta r_i^* + X_i'\gamma_0 + \epsilon_i \tag{11}$$

where Δr_i^* is the empirical likelihood a helmet will save one's life, estimated from ridership and helmet efficacy. This is similar to León and Miguel (2017) and Ito and Zhang (2020) which esti-

[20] I pre-specified two sets of IVs, but the original pre-analysis plan (PAP) registered a slightly different version of the interacted IV using n_i, motorcycle trips/week, instead of $r_{0,i}$. This inaccurately assumed n_i predicts $r_{0,i}$. An amendment before wave 2 specified $r_{0,i}$.

[21] The initial PAP only listed the demographic variables which produces similar results (Appendix Table A8). The PAP amendment specified the full set of potential controls and enumerator FEs.

[22] Heteroskedasticity would arise mainly through VSL heterogeneity, whose contribution appears small: the standard deviation of WTP is roughly 2.9 times VSL times the perceived likelihood a helmet saves one's life. Over-identified confidence sets use CLR inversion (Moreira, 2003) and just-identified models use AR inversion (Anderson and Rubin, 1949) for efficiency.

mate logit models under rational expectations. The specification is linear since WTP is measured. Second, I estimate

$$VSL_i = \frac{v_i}{\Delta r_i^*} \tag{12}$$

This assumes rational expectations and that agents do not receive utility from any characteristics of helmets other than safety. This specification follows Berry et al. (2020) and is similar to Kremer et al. (2011). The method is typically used with health products, where assumptions are plausible.

These approaches do not separate willingness to pay for morbidity prevention from mortality prevention, so estimates are upper bounds on VSL (León and Miguel, 2017; Kremer et al., 2011).

5 Results

5.1 Estimates of the value of a statistical life

Elicited beliefs about the probability of a fatal motorcycle accident fall in plausible ranges and are consistent across measures, supporting their use in estimating VSL. The mean reported 5-year mortality risk is 0.034, with a median of 0.001.[23] For comparison, the estimated median empirical risk is about 1/4,000. Beliefs about deaths/10,000 passengers and one's own risk are significantly correlated ($R^2 > 0.1$), though agents perceive lower personal versus average mortality risk.

Beliefs about 1-year and 5-year mortality risk are also consistent ($R^2 = 0.57$). Over 60% of respondents report 5-year risk estimates between 4 and 6 times their 1-year estimate, and less than 1% gave logically inconsistent answers. Respondents' 5-year risk beliefs are also aligned with per-trip fatality risk assessments ($R^2 = 0.2$).

First stage: Effect of information on beliefs

Table 2 indicates that randomized information exposure had a statistically significant effect on the agents' beliefs that a motorcycle helmet will save their life.

Respondents were first presented with estimates of their unhelmeted mortality risk constructed using data from the NTSA. Consistent with anecdotal reports of mistrust in the NTSA, respondents reported no change in beliefs on average given this information (columns 1 and 2). There is no significant difference between either treatment arm, which received the same information, and the control. This suggests agents did not feel compelled (e.g. by demand effects) to misreport beliefs.

[23] Results are robust to excluding implausible responses or winsorizing beliefs (Appendix Table A3).

Respondents did update beliefs about the effectiveness of helmets in response to the treatments (Table 2, columns 3-4), which provides the primary variation used by the interacted instrument. At baseline, agents overestimated the effectiveness of helmets: the mean control group belief was 79%. Exposure to the 70% ("high") and 42% ("low") effectiveness estimates led respondents to revise beliefs downward to 75% and 65%, respectively.[24]

The change in beliefs about helmet effectiveness was large enough to shift respondents' perceived probability that a helmet would save their life (the product of accident risk and helmet efficacy). The dependent variable in Columns 5-6 of Table 2 equals 10,000 times this perceived probability. Relative to the control group, the low-treatment group reports values that are about 80 units lower on average (29%, $p < 0.01$). The high-treatment group does not differ significantly from control (the signal was close to priors) though point estimates move in the predicted direction.

The value of a statistical life

Primary estimates of the value of a statistical life are reported in Table 3. The preferred estimate in column 1 is $523, suggesting low demand for mortality risk reduction. A weak-instrument robust 95% confidence set excludes values below $214 and above $1,043. Estimates are consistent across specifications, both with and without controls, and across instrument sets.

Column 4 instruments for risk using an observational shifter of beliefs: an indicator for knowing a motorcycle accident victim, conditioning on taxi terminal fixed effects. The resulting estimate, $560, is close to and statistically indistinguishable from the experimental estimates. The identifying logic is that, among respondents in the same area, whether someone they know has been in an accident is partly a matter of luck. The estimate is nonetheless suggestive, since victim exposure may be endogenous. The aim of this exercise is not to replace information treatments. Its value is instead that this variation cannot be driven by demand effects, and that it moves beliefs and WTP by statistically significant margins, suggesting that beliefs affect real outcomes.

The estimates pass a battery of robustness checks. Columns 1-3 of panel b report interacted IV estimates leaving out one experimental arm, testing if results are stable shifting beliefs to different parts of the distribution. These results control for fixed effects for the taxi terminal where the trip originated to improve first stage power to overcome the reduction in sample size. The estimated

[24]Under Bayesian updating one would not expect a 28 p.p. gap in beliefs between the treatment arms since beliefs are a weighted average of priors and the signal. Estimates both imply a weight of about 60% on priors.

VSLs are similar to the primary estimates, though the first stage is weak when the control arm is excluded, leaving a confidence set that extends to $5,160. Column 4 reports pre-specified treatment only estimates dropping the control arm. The result is statistically indistinguishable from panel a.

Appendix Table A3 winsorizes beliefs about helmet efficacy and willingness to pay. VSL estimates are broadly similar, though some rise because winsorizing shrinks the denominator. Appendix Table A4 shows that results are statistically indistinguishable, and remain low, accounting for changes in planned ridership (Panel a), beliefs about helmet lifespan (Panel b), and selection into ridership volume (Panel c).[25] IV probit results regressing helmet receipt on price (BDM draw) and perceived risk reduction (instrumented by treatment) yields a VSL estimate of $162 (Appendix Table A5), indicating that the low VSL is not driven by valuations above the maximum BDM draw.

As detailed in section 5.2, the local average VSL identified weights individuals with higher priors about the riskiness of motorcycles more heavily. Appendix Table A6 asks whether priors are correlated with observable respondent characteristics, which would indicate that the parameter is identified over a selected population. Only gender is significantly correlated, which I show below does not significantly predict VSL. The similarity of the treatment only and interacted IVs also support the view that a particularly selected sample is not likely driving the estimates, as the interacted instrument weights those that perceive helmets to be riskier more.

Although demand for safety in this setting appears low, economic theory suggests these estimates are consistent with much higher VSLs in richer settings, where the opportunity cost of funds is far lower. Comparing results from this paper to estimates across contexts reveals that this study's estimates are of a similar order of magnitude to those from similar income levels. Some differences may be due to methodology, but this helps to confirm the importance of income.

Figure 1 plots recent LMIC revealed preference estimates of VSL, adjusting for inflation between publication years.[26] I also include Greenberg et al. (2021), which examines US soldiers, as a rich country comparison. The estimates here fall well below those from wealthier populations but are close to those from populations at comparable income levels. The only other Kenyan estimate, Kremer et al. (2011), lies within the 95% confidence set of the preferred estimate. Berry et al.

[25]Estimates exclude 35 observations from the second day of data collection, as specified in the PAP amendment, because in some cases motorcycle drivers were pretending to be passengers to obtain a free helmet and submitting false surveys. Estimates are similar if they are included (Appendix Table A8).

[26]I exclude wage-risk studies because of evidence that labor markets do not always clear in developing countries (Breza et al., 2021), but I include all other LMIC revealed preference estimates of VSL that I am aware of.

(2020) report a median VSL just over $4,000 in Ghana based on treatment effects after one month. The authors also report a median willingness to pay to avert a case of child diarrhea of USD -0.14 based on one year treatment effects, but do not convert this into a VSL since it would imply a negative value. For comparability, I convert this estimate into a VSL using the same approach the authors apply to the one-month estimate. Ito and Zhang (2020) and León and Miguel (2017) find much higher values in wealthier LMIC populations. Shrestha (2020) reports a higher value among Nepalese migrants, over $300,000, though its confidence interval stretches well below $0.

As a thought experiment, one may assume that one's utility from being alive is constant across income levels and calculate what curvature of marginal utility of consumption would be required to rationalize differences in VSL. Under a CRRA utility function, the gap between this study's estimate and the VSL of $700,000 in León and Miguel (2017) implies a coefficient of relative risk aversion of $\theta = 3.01$.[27] This falls solidly in the range of empirical estimates: Havránek (2015) reports a mean elasticity of intertemporal substitution of $1/3$ in a meta-analysis, equivalent to $\theta = 3$. While CRRA utility may not hold exactly, and methodological differences may drive gaps in results, this exercise demonstrates that large cross-country variation in VSL can readily be rationalized by differences in the marginal utility of consumption. This is consistent with Hall and Jones (2007), which finds that health investment rises sharply with income.

Heterogeneity in the value of a statistical life

If VSL is heterogeneous, estimates from this population may differ from the average VSL relevant to other policies. This paper studies a group of direct interest for transportation policy. To assess broader applicability, I examine how VSL varies with observable characteristics in Table 4. The patterns match theoretical predictions, but the estimated heterogeneity is modest, so the conclusion that demand for safety is low seems unlikely to change across similar populations.

I make two econometric changes for power. First, I pool high treatment and control observations, where posteriors are similar (Table 2), to limit small cell sizes within heterogeneous groups. Second, I use an interacted first stage with $r_{0,i}$ set to a respondent's prior about their own mortality risk. As discussed in Section 4.2, this risks attenuating estimates if measurement error in $r_{0,i}$ is strongly correlated with that in Δr_i. VSL estimates fall to $224 with this estimator, though they

[27] Assume $\beta_i = \beta, \alpha_i = Y_i^{-\theta}$ where Y_i is income. Then $\frac{VSL_1}{VSL_2} = \left(\frac{Y_1}{Y_2}\right)^{\theta} \Rightarrow \theta = \log\left(\frac{VSL_1}{VSL_2}\right) / \log\left(\frac{Y_1}{Y_2}\right)$. Thus $\theta \approx \log(700,000/523) \div \log(75,000/6,866) = 3.01$

are not significantly different. I use this instrument because it increases first stage power, making heterogeneity analysis feasible. Measurement error would most likely bias estimates towards 0, allowing one to draw inferences from the results, although they should be interpreted cautiously.

I first examine heterogeneity by age (Table 4, column 1). I estimate similar VSLs below and above the median value. This result aligns with prior empirical work studying VSL over the life-cycle, which finds that the parameter peaks near age 40 (Aldy and Viscusi, 2007).

Estimates of VSL heterogeneity by income show a strong positive relationship, matching theory. I consider heterogeneity with respect to asinh(wages) in Table 4, column 2, demeaned so that the first coefficient captures average VSL.[28] Estimates indicate that a 1% increase in wages is associated with an increase in VSL near $4. This implies a back of the envelope income-elasticity of 1.14. This value is particularly striking because income is measured using a short question likely to attenuate the elasticity. Thus the estimate of 1.14 likely represents a lower bound, supporting the view that VSL increases sharply with income. VSL estimates are also about $500 higher among those with above-median income (Table 4, column 3) and reported health (Table 4, column 4).

In contrast, there is no significant heterogeneity with respect to having children, digit span recall (a cognitive test of memory), education, or gender. This suggests that the fact the study sampled more men than women is not likely to substantially influence estimates.

5.2 Are VSL estimates biased by non-classical measurement error?

The primary identifying assumption in this paper is that any gap between agents' true subjective beliefs and their reported beliefs reflects only classical measurement error. Research shows that agents' decision-relevant beliefs can be shaped by heuristics like rounding (Lacetera et al., 2012). This renders common tests of belief accuracy (e.g. bunching analysis) uninformative, since the method allows for arbitrary differences between subjective beliefs and the empirical truth. To address this, the design incorporates features that facilitate tests of belief validity.

I included two treatments of different intensity so that VSL may be estimated at multiple points of the belief distribution. Estimates should be invariant to the arms included under classical error. However, if beliefs were misreported – unless misstatement was proportional across arms – es-

[28] I set the wages of the unemployed to 0 and use asinh since unemployment is often involuntary. Results are similar, but less precise, dropping the unemployed and using log(wage).

timates would differ. For example, if lower reported helmet efficacy in the low treatment arm reflected demand effects, estimates using the low treatment and control groups would be smaller (since the denominator would be over-estimated) than those using the high treatment and control. Estimates are similar across all combinations of arms (Table 3), helping rule out this concern.

The two instrument sets facilitate a second test. The interacted IV places greater weight on individuals with high priors about motorcycle risk. If reported beliefs were subject to non-classical error correlated with true beliefs, the IVs would yield different estimates. Appendix B presents a formal argument, based on the literature on local average treatment effects (e.g. Abadie et al., 2023; Mogstad et al., 2018), and shows that this analysis can test for common patterns of misreporting.

Intuitively, willingness to pay for a helmet is a function of an agent's true belief about motorcycle risks, $r_{0,i}$, helmet effectiveness, and their VSL. Treatment only estimates are proportional to the covariance between willingness to pay and treatment assignment, so a weighted average of VSLs proportional to $r_{0,i}$ is identified. With the interacted instrument, first stage predictions also depend on elicited $r_{0,i}$. This leads to a weighted average of VSLs proportional to $r_{0,i}^2$ if errors are classical. If beliefs are systematically mismeasured, weights will be a function of the correlation between $r_{0,i}$ and error with this instrument, whereas treatment only weights will remain proportional to $r_{0,i}$. Thus, the IVs will produce different estimates, unless errors are orthogonal to true beliefs.

For instance, some agents report very high risks of riding motorcycles. Are they overstating their belief? This would result in the interacted IV producing a lower estimate, as long as those reporting very high beliefs have above average true expectations. The appendix argues that typical forms of rounding or misreporting would also cause the IVs to diverge.[29] Yet empirically, VSL estimates are similar across instrument sets, and a Hansen J-test fails to reject their equality ($p > 0.5$), suggesting that significant bias from misreported beliefs is unlikely.

Demand effects are further ruled out by reduced willingness to pay for a helmet in the low treatment arm (Appendix Table A7), providing revealed preference evidence of belief updating. Low treatment respondents also moved towards the signal without adopting it: only 1.3% report the 42% figure. If beliefs are decision-relevant, one would expect the fall in WTP to be concentrated among those who view motorcycles as risky. There is suggestive evidence supporting this view. Low treatment assignment reduces helmet demand by almost $6 among respondents who report a

[29] Appendix Figure A4 illustrates this in the case of rounding.

prior risk of a fatal accident of at least .01 (at this point a VSL of $523 predicts an effect of about $1). Below this cutoff, the point estimate is negative but small and insignificant (the difference between groups is not significant). Helmet demand is also similar in the pure control and control arms (Appendix Figure A7), even though the control group was asked about motorcycle risk.

To further address concerns about difficulty with small probabilities, I proxy for perceived risk reduction by combining beliefs about the number of motorcycle deaths/10,000 users over five years with reported helmet efficacy, a format that permits an integer response. I estimate a VSL of $145 (Appendix Table A8), below the preferred estimate, which is the opposite change one would expect if difficulties with low probabilities suppressed estimates. Appendix Table A8 also reports estimates that top code beliefs at the 90th percentile to limit sensitivity to the right tail. Point estimates increase modestly, to $654 with the treatment only IV and $1,032 with the interacted instrument, but remain statistically indistinguishable from results using the direct belief data.

5.3 Testing rational expectations

The primary assumption relaxed by this paper is that agents' beliefs satisfy rational expectations. Results suggest that this assumption is violated in this context.

This study captures both agents' subjective beliefs and their empirical mortality risk, allowing for tests of rational expectations. The two variables are uncorrelated ($\rho = -0.0008$) and respondents report average beliefs above empirical estimates, suggesting a violation of rational expectations. However, measurement error could obscure a relationship. I therefore instrument for final beliefs using another measure (deaths/10,000 users), which addresses bias from independent classical errors (Gillen et al., 2019). This yields a significant positive relationship, but the coefficient is below 0.001. I reject equality to 1 (the rational expectations benchmark) with 99.9% confidence. Because this test may be sensitive to correlated errors across belief measures, I supplement the analysis with two additional data sources to examine how agents form expectations.

First, the survey asked respondents the information sources they used to form beliefs, plotted in Appendix Figure A5. The most common source is own experience (79%), followed by family members (48%) and social media (38%). Under a third of respondents reported consulting more objective media or government sources. These information sources are consistent with deviations from rational expectations since they depend on private information and are prone to bias.

Second, I regress beliefs on (1) variables that shift empirical risk and (2) variables that should influence beliefs (but not necessarily true risk) if reported information sources are accurate. I use trip volume and duration as empirical shifters, since more frequent motorcycle use mechanically increases risk, particularly conditional on taxi terminal fixed effects. In fact, a 1% increase in ridership is associated with a 0.13 percentage point increase in the likelihood of reporting a prior accident in the data ($p < .01$). To test if agents form beliefs based on personal or social network experiences, I consider indicators for having suffered a motorcycle accident or knowing a victim.

If rational expectations held, one would expect positive coefficients on the empirical shifters of risk. Table 5 demonstrates that this is not the case: beliefs about risk have no statistically significant relationship with trip volume or the length of a typical trip with or without terminal fixed effects and covariates. In contrast, accident history (personal or in one's social network) is strongly predictive of beliefs. These patterns indicate a violation of rational expectations.

One remaining concern is that accident history may reflect risk heterogeneity observable to respondents but not to the econometrician, in which case rational expectations could hold under a more complete empirical model. At a minimum, the results highlight the sensitivity of rational expectations to the researcher's data. Moreover, I test whether reported five-year risks vary with rainfall on the survey day. Weather today should not affect long-run accident risk, but rain heightens the salience of motorcycle dangers. Respondents reported roughly twice the risk if it rained, indicating a departure from rational expectations (Table 5, Column 5).

I next examine if typical approaches to estimate VSL, which assume rational expectations, consistently recover the parameter in Table 6.[30] Estimates from typical methods fall outside the experimental confidence sets, and different observational approaches yield starkly different results.

Columns 1-4 report estimates of VSL from Equation 11. Column 1 omits covariates; columns 2-4 use double-post LASSO for covariate selection, with column 3 adding enumerator and column 4 taxi terminal fixed effects. Estimates are sensitive to controls and confidence intervals are wide. The point estimate exceeds $15,000 without controls and is negative in column 3. Columns 2 and 4 yield point estimates closer to the experimental benchmark, around $2,700 and $1,500, respectively. While this offers some evidence that rational expectations-based methods can recover

[30]One must take a stance on the efficacy of helmets. Panel a reports results using Liu et al. (2008). Panel b reports estimates using Ouellet and Kasantikul (2006). Discussion focuses on panel a.

VSL, these results may simply reflect a weak correlation between willingness to pay and ridership. Estimates using Equation 12 are reported in column 5: the mean exceeds $380,000.

The experimental design relaxes rational expectations and addresses endogeneity at once, so the gap between it and conventional estimates cannot be attributed to either alone. Column 6 separates the two. Like the experimental estimates it uses subjective beliefs, but it instruments for final beliefs with priors about deaths per 10,000 passengers over five years rather than with randomized information. If measurement error is independent across the two elicitations, this corrects attenuation while leaving omitted variable bias intact. The resulting estimate of about $110 lies below the experimental value, suggesting that endogeneity also binds in this setting.

5.4 Other threats to identification

There are several threats to the experimental VSL estimates aside from misreported beliefs.

First, the treatments may have influenced beliefs about dimensions of helmets other than their life-saving potential, particularly injury prevention. To limit this, the information treatments excluded injury content and no injury-related questions were asked. If injury beliefs did shift, the VSL estimates would be upper bounds so long as perceptions of helmet efficacy at preventing injury moved in the same direction as consumers' beliefs about their efficacy at preventing death. An upper bound does not threaten the conclusion that revealed preferences are inconsistent with the values of VSL used in policy. This limitation is also shared by other revealed preference estimates of VSL: because the choices they study rarely vary mortality and morbidity risk independently, authors often present their estimates as bounds (León and Miguel, 2017; Kremer et al., 2011).

Second, VSL may be underestimated if agents did not plan to regularly use helmets. Piloting indicated that (consistent with the fact helmets are costly) respondents believed that they would regularly use them. Because anticipated rather than realized use is the decision-relevant parameter, such bias seems unlikely. Low anticipated use would itself be hard to reconcile with a high VSL, since wearing a helmet one owns costs little beyond inconvenience. Resale is also improbable because respondents all accepted a helmet with quality tags and the box removed, which undermines resale value but avoids hassle if worn. Even if respondents anticipated using a helmet 1% of the time, VSL estimates would remain below those used in policy. Appendix Table A4 further shows that estimates are similar when adjusting for anticipated duration of use.

Monotonicity violations could affect the interpretation of estimates as an average VSL. But only 3% of control respondents had priors below the 42% figure presented to the low treatment group, and estimates are similar when only control and low treatment observations are considered.

Finally, respondents could face capital constraints. Demand was elicited as a choice between a helmet and cash, so valuations were not limited by cash on hand. Liquidity needs may remain, but median willingness to pay is only a quarter of weekly wages, and estimates rise to about $678 when the unemployed, the most constrained respondents, are excluded (Appendix Table A8). In any case, these estimates reflect demand under the constraints consumers actually face, the relevant policy benchmark, unless liquidity happened to be unusually tight during the study.

It remains possible that these estimates are biased, for instance if beliefs are misreported in a way my tests cannot detect. A distinct concern is that behavioral biases suppress demand for preventive health, so that revealed preference need not recover welfare-relevant valuations; I return to this in the next section. But the results show that revealed preference VSL estimates remain low when leading threats to identification are relaxed, using a method distinct from those in the existing literature, indicating a mismatch between empirical evidence and policy assumptions.

6 Discussion

This section examines whether the VSL estimates in this paper are plausible and considers potential limitations of the estimates and with leveraging VSL to value health investments in poor settings.

6.1 Is an average VSL below annual income plausible?

The average VSL estimated in this study is less than annual income. Is this value plausible?

VSL is not the present value of lifetime consumption. The parameter trades certain costs today against uncertain future safety benefits. It also reflects expected utility from being alive minus expected utility from not being alive, and the latter term may be important in some cultural and religious contexts. Someone who believes in an afterlife, for instance, may expect considerable utility after death. Empirical evidence supports this: VSL falls sharply with the strength of religious (de Lima, 2020) and fatalistic beliefs (León and Miguel, 2017). This may help explain the low VSL in Kenya, which has one of the highest rates of belief in an afterlife globally (Evans et al., 2025) and where fatalism has been shown to limit health investments (Oo et al., 2024). This

motivates the comparison of this study to León and Miguel (2017) when examining cross-country income comparisons, as Sierra Leone also has pervasive fatalistic and religious beliefs.

Of course, these are hypotheses rather than findings, offered mainly to show that a low VSL should not be dismissed out of hand. The study was not designed to test why VSL estimates are low, but to evaluate a method for estimating the parameter. Why revealed preference estimates are often low in LMICs is an open question (Jack and Ryan, 2026), in part because few VSL estimates exist and their methods vary, making it hard to test the role of features like discounting or religion. The method here could help address this gap by generating comparable estimates across settings.

Aside from conceptual arguments, one may question whether low VSLs are empirically consistent with health investments in related settings. In fact, the results align well with evidence of low demand for preventative health products in LMICs (Dupas and Miguel, 2017), which offers a particularly close comparison since such products present a tradeoff between upfront costs and uncertain future health benefits. Because this evidence comes from realized purchases rather than VSL estimation, it is independent of the method introduced here.

Cohen and Dupas (2010) show that 40% of pregnant women in Kenya pay $0.60 for a bed net that all adopt for free. Malaria nets save an estimated 5.6 lives/1,000 users, implying a cost per life saved below $110 at $0.60 (Pryce et al., 2018). Uptake of deworming pills falls by 60 percentage points when costs rise by $0.30 (Kremer and Miguel, 2007). And Banerjee et al. (2010) show that incentives worth $4 for vaccinating children (at no cost) increase uptake from 18% to 39%.

These results indicate that the VSL estimate captures a real signal about behavior. But a common set of constraints or decision-making biases could depress both preventive health demand and measured VSL. Some evidence points against particular candidates: voucher experiments that shift the timing of payments are hard to reconcile with present bias (Dupas and Miguel, 2017). Others cannot be ruled out, and would contribute to low estimates here and throughout the literature. What the method does establish is that low VSL estimates are not an artifact of proxying subjective beliefs with empirical risk or omitted variable bias, two leading concerns raised in the literature.

6.2 Welfare and policy implications

This section considers the policy implications of the estimated VSL. First, I apply the estimate within prevailing benefit-cost practices, which implies that reallocating funds from health programs

towards poverty alleviation would raise welfare. Second, I argue that this conclusion may reveal a limitation of that practice, not optimal policy. Donor-funded health programs often fall outside the theoretical scope of VSL, causing benefit-cost analyses to understate welfare gains.

Implications under prevailing benefit-cost practices

The typical approach to valuing lives saved in benefit-cost analyses conducted in LMICs is to take a high-income country VSL estimate, then assume an elasticity of VSL with respect to income to estimate the VSL of the population studied (Robinson et al., 2019). The elasticities imposed are often near one, producing values substantially higher than the estimates in this paper. These results suggest that this practice poorly captures the preferences of affected populations.

To assess how the estimates affect policy conclusions, I selected five benefit-cost analyses of Kenyan programs.[31] I re-calculate benefit-cost ratios (BCRs) using this paper's VSL and compare them to the original findings. Appendix Figure A3 shows that BCRs fall by over 99% on average. In four cases, the BCR crosses 1, reversing the policy conclusion. The populations evaluated in these papers may differ from this paper's sample. But the exercise illustrates that many health programs appear far less cost effective when lives saved are valued at locally estimated VSLs.

The same mismatch appears in practice. GiveWell, an NGO that has matched over $1 billion to charities, weights averting the death of someone aged 15-49 at 104 times doubling their consumption for a year,[32] directly affecting which charities are recommended. Under the estimates in this paper, such weights understate the value of consumption gains by orders of magnitude, implying that shifting funding towards programs like cash transfers would raise recipient welfare.

Why a low VSL does not imply health funding should fall

A natural objection to these conclusions is that health programs are widely regarded as valuable, so a benefit-cost calculation implying that such funding should be cut may indicate a flawed calculation rather than a flawed program. One possible explanation — compatible with this view and the low VSL — is that benefit-cost analysis may apply the parameter outside the scope of decisions for which it is informative. The empirical results may therefore indicate that assumptions used to apply VSL in benefit-cost analysis are flawed, not that healthcare is significantly

[31] Accessed June 2024. These are Babagoli et al. (2022) (which reports separate ratios for menstrual cups and sanitary pads), Mwai et al. (2023) (primary health), Hamze et al. (2017) (cleft lip/palate repair), and Oyugi et al. (2023) (maternity care), yielding five benefit-cost ratios from four papers.

[32] Based on the 2023 update, which draws on stated-preference VSL elicitation from (Redfern et al., 2019).

over-funded. Examining how VSL is actually used in academic benefit-cost analyses and policy guidance supports this interpretation: the parameter is frequently applied to decisions it cannot reliably price, in ways that understate the benefits of programs serving poor populations.

VSL captures individuals' willingness to trade off marginal future changes in their mortality risk against their own consumption today. This can inform many important policy decisions, such as how much a government should tax its population to abate pollution or improve safety infrastructure. But there are three classes of decisions where the framework does not recover welfare.

First, VSL under-values policies that affect mortality risk non-marginally. An individual with a low VSL may have a high willingness to pay to avert certain death today. Yet in practice, VSL is routinely used to price lives saved by policies affecting risk well outside the marginal range, including treatment for conditions such as AIDS where the risk reduction is large and immediate.

Second, VSL captures only preferences for mortality risk reduction, not reduced illness or disability. Such benefits are typically indexed to VSL through disability adjustments, which assume that the value of these attributes relative to mortality risk reduction is constant across income levels. But theory predicts that the relative importance of morbidity to mortality should rise as income falls, since coping with a negative shock is more challenging with few resources. So disability-adjusted valuations anchored to VSL may understate program benefits in poor settings. Helmet demand in this study illustrates the point: valuations often exceeded the wholesale price despite the low VSL, and respondents in piloting frequently cited protection from non-fatal injuries and the resulting financial costs as important reasons they wanted one. In addition, features like fatalism or religiosity, which may drive down demand for mortality risk reduction specifically, may create heterogeneity in the relative importance of morbidity prevention.

Third, and most directly related to developing countries, VSL recovers welfare only when a fixed population faces a tradeoff between mortality risk and consumption. This framework is often appropriate for domestic decisions. But a large share of health spending in LMICs is externally financed by global health institutions or philanthropic foundations. In such cases, valuing lives saved at the recipient population's VSL substantially understates welfare gains.

To illustrate this, suppose there are otherwise homogeneous high and low-income consumers indexed by h, ℓ. As in Section 3, let α denote the marginal utility of consumption and β the expected utility from averting death. Suppose a high-income government distributes N vaccines to

low-income consumers, each of which saves a life with probability p. The cost is $\zeta \cdot N$, paid for by N high-income consumers. The global (utilitarian) welfare gain is

$$W = N \cdot p \cdot \beta_\ell - N \cdot \zeta \cdot \alpha_h$$

W is positive if $p \cdot VSL_\ell \cdot \frac{\alpha_\ell}{\alpha_h} > \zeta$. Valuing lives saved at VSL_ℓ understates monetized benefits by $\frac{\alpha_\ell}{\alpha_h}$. Under a CRRA parameter of 3, a tenfold donor-recipient income gap gives $\frac{\alpha_\ell}{\alpha_h} \approx 1,000$.

How common are such applications?

The application of VSL to externally funded development programs is widespread in policy and academic research. Institutions including the WHO (WHO, 2001), World Bank (Wijnen et al., 2025), OECD (OECD, 2025), and US EPA (United States Environmental Protection Agency, 2023) explicitly value lives saved at local VSLs in benefit-cost analysis or index cost effectiveness thresholds to local income.

Academic studies follow this practice, often because they use benefit-cost procedures from policy. I used a large language model to classify applications of VSL in NBER working papers since 2000, then manually checked cases where the model reported a mismatch between an LMIC program's donor and recipient.[33] Among VSL applications to LMICs, 40% value lives saved by programs financed by NGOs or other external entities, or evaluate policies such as pollution abatement whose costs fall outside the recipient population. In almost all such cases, the recipient VSL is applied. Some classifications involve judgment (e.g. programs funded by an NGO that a local government could implement), so these exact values may be sensitive to a reader's interpretation.

Previous research has raised related concerns (Sunstein, 2013). The practice persists in part because no tractable alternative allows policies to be compared on a common basis (OECD, 2025), and because proposed substitutes require inputs that are rarely available, such as each population's marginal utility of consumption (Adler, 2016).

The case for using donor VSLs in benefit-cost analysis

I argue that a simple change to benefit-cost analysis—using donor rather than recipient VSLs to value lives saved by externally financed programs—would likely raise welfare without adding to the complexity of the exercise. Differentiating equation 4 gives

$$\eta_{vsl,c} = \eta_{\beta,c} - \eta_{\alpha,c}$$

[33] Details, including the classification prompt and classified papers, are available in appendix C.

where $\eta_{vsl,c}$ is the elasticity of VSL with respect to income, $\eta_{\beta,c}$ that of the utility gain from surviving versus not, and $\eta_{\alpha,c}$ that of the marginal utility of consumption. Since utility from being alive is concave in consumption, typically $0 < \eta_{\beta,c} < 1$, while macroeconomic estimates place $\eta_{\alpha,c} < -1$. Most of the growth of VSL with income therefore comes from the marginal utility of consumption rather than from the utility of survival, consistent with the calibration in section 5.1.

This indicates that benefit-cost ratios computed with the donor VSL are typically closer to the truth than those computed with the recipient VSL. Let R be the ratio of donor to recipient income and r^* be a program's true benefit-cost ratio, so that a utilitarian social planner implements the program if $r^* > 1$. The donor rule funds programs with r^* between $R^{-\eta_{\beta,c}}$ and one, so at worst it funds a program returning only a fraction of its costs. The recipient rule rejects programs with r^* between one and $R^{|\eta_{\alpha,c}|}$, which in the example above means benefits up to 1,000 times their cost. Appendix D formalizes this: when the donor is at least twice as rich, the donor rule's worst-case loss is bounded by program costs while the recipient rule's can exceed them many times over, so that minimax regret is lower with the donor rule. The donor rule also yields higher expected welfare unless an unusually large share of programs fall in its narrower error range. Similar logic applies to ranking programs where $r^* > 1$ across recipient countries, for instance in deciding which programs to scale as is commonly faced in development policy.

Local VSLs are the correct input for decisions in which a donor allocates a fixed budget *within* a recipient setting. A funder choosing between a health program and a cash transfer for the same population confronts the mortality risk-consumption tradeoff that VSL prices. Misallocation arises when local VSLs are instead used to compare programs *across* populations of different income levels, in which case cross-country variation in VSL reflecting tighter budget constraints among the poor is attributed to a lower value of averting death.

6.3 Extending the method to other non-market amenities

The method for estimating demand for non-market goods introduced in this paper could also be applied to value other amenities. I apply the method—updating beliefs about the effect of a product on exposure to a non-market amenity, then eliciting demand for the product—to VSL because the parameter is important for policy and because the design addresses leading concerns with existing estimates. But the approach could be readily adapted to other goods.

Applying the method requires that a product exists which clearly affects exposure to the amenity of interest, that the product has not been ubiquitously adopted, and that there is credible information about the relationship between the product and the amenity that can be used to update beliefs. These conditions may be challenging to satisfy in high-income settings where product diffusion is rapid, although strong cultural forces may produce informative populations in some cases (e.g. bicycle helmet usage is rare in the Netherlands). In developing countries, the slow diffusion of products creates more opportunities to apply such tools.

I note that the approach does not require that a product does not meaningfully affect other attributes of one's utility function, for instance in this setting helmets also affect comfort and morbidity risk. However, other attributes can matter if a signal about the targeted relationship also causes beliefs about other dimensions to update. In this setting, signals about effectiveness of helmets at preventing mortality could plausibly affect beliefs about morbidity risk, but a clear bound is identified. In settings where a bound is not informative, one can implement a simple extension of the model where they independently update beliefs about the nuisance parameter to control for it in the model.

One application is reducing the dependence of benefit-cost analysis on VSL. Health benefits are often indexed to VSL through disability adjustments, in part due to the difficulty of valuing non-fatal health states. This practice may understate program benefits by more in poorer settings if coping with serious illness or disability is more challenging without financial resources or a social safety net. Directly estimating demand for averting morbidity therefore appears to be a natural next step, and could be done by updating beliefs about the effectiveness of helmets, water chlorination, or malaria nets at preventing illness or disability. The approach could also be valuable for studying preferences for environmental quality. For instance, one could apply the method to air purifiers to value clean air, or to water filters to value reductions in waterborne disease risk.

7 Conclusion

This study introduces a framework for experimentally estimating demand for non-market amenities using subjective belief data and uses the approach to estimate the value of a statistical life. I leverage the fact that products exist which affect an agent's exposure to non-market goods. This paper demonstrates that researchers may cause agents to update their beliefs about how such a

product affects the amenity of interest, then examine how product demand changes to uncover willingness to pay for the amenity. This method is tractable and low-cost in many settings where exogenous variation in attribute exposure does not exist, and it avoids the need to assume that agents' beliefs satisfy rational expectations, which appears to be violated in this setting.

I estimate low demand for mortality risk reduction in Kenya. This is consistent with a common finding that uptake of preventative health products is low in LMICs. Two implications for policy follow, and they point in opposite directions. First, when a population faces a tradeoff between its own consumption and its own mortality risk — locally financed public investment, or a donor allocating a fixed budget for a given population between health and cash — these estimates are the relevant ones, and they are far below the values currently in use. Second, current practice values lives saved from externally funded health programs or pollution abatement at the recipient country's VSL. This understates the welfare gain by the ratio of recipient to donor marginal utility of consumption. I argue that valuing lives saved at donor country VSLs would likely raise welfare, since differences in the marginal utility of consumption across income levels dominate differences in utility from survival given standard estimates of the curvature of consumption utility.

Finally, the method may help value non-market goods beyond mortality risk. Willingness to pay to avoid disability, illness, or pollution is often estimated from unincentivized survey responses. This paper offers an inexpensive revealed preference alternative based on product purchase decisions. Three applications seem especially promising. First, within mortality-risk valuation, the framework could be combined with experimental tests of present bias, or applied across settings to study the sources of VSL heterogeneity. Second, it could value morbidity prevention, reducing the dependence of benefit-cost analysis on VSL. Third, it could provide revealed preference measures of demand for environmental quality. In each case, a lack of exogenous variation in amenity exposure has created a barrier to producing revealed-preference estimates. My design produces comparable identifying variation in expected utility by shifting beliefs about a product, lowering the cost of incentivized amenity estimates.

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Figure 1: Revealed preference VSL estimates across studies

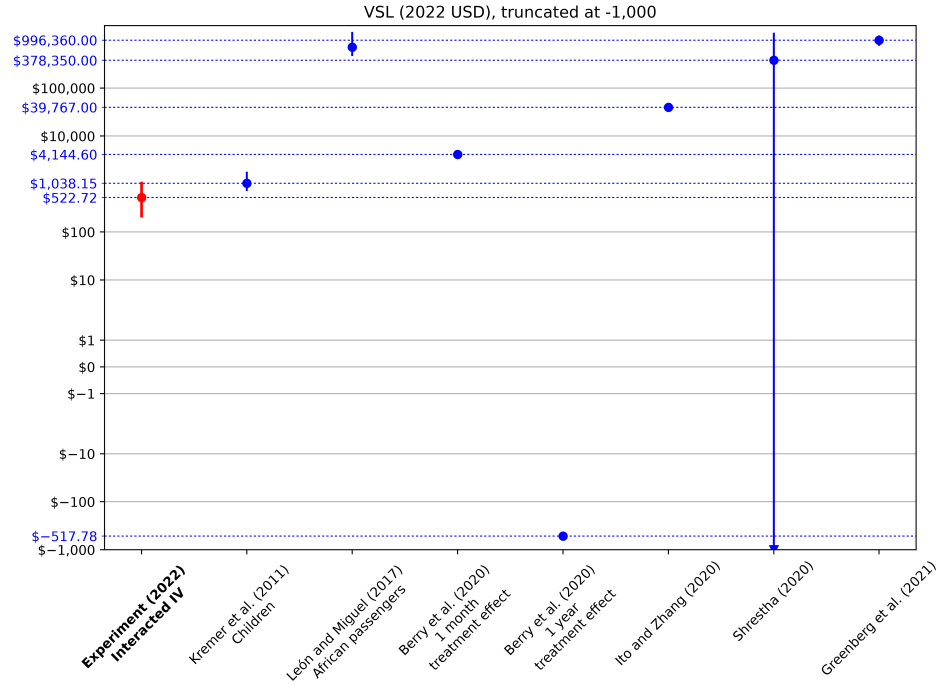


Figure 1 plots revealed-preference VSL estimates and (where available) 95% confidence intervals from this paper, Kremer et al. (2011, rural Kenya), León and Miguel (2017, wealthy international travelers in Sierra Leone), Berry et al. (2020, rural Ghana), Ito and Zhang (2020, urban China), Shrestha (2020, potential Nepalese migrants), and Greenberg et al. (2021, American soldiers). Greenberg et al. (2021) is included for comparison to a high-income setting. The other estimates are from low and middle income economies. Berry et al. (2020) present the 1 month figure in their paper, and report median WTP to avert a case of child diarrhea based on 1 year treatment effects but do not scale it into VSL. I convert this to a VSL using the same approach they apply to 1 month estimates in their paper, but I note this was constructed here and not taken directly. All estimates are presented in 2022 USD calculated by inflating based on the paper's publication year using the CPI inflation calculator. The lower bound of the 95% confidence interval from Shrestha (2020), which is below -$600,000, is truncated at -$1,000. Ito and Zhang (2020) does not report a confidence interval.

Table 1: Summary statistics and balance: Demographics

	(1) Control	(2) Pure control - control	(3) Low treatment - Control	(4) High treatment - Control	(5) High - low treatment
Age	32.735	0.105	0.282	0.659	0.377
	[8.126]	(1.159)	(0.572)	(0.589)	(0.590)
Female	0.352	-0.039	-0.064**	-0.050	0.014
	[0.478]	(0.060)	(0.030)	(0.030)	(0.029)
Health (1-5)	3.403	-0.034	-0.056	-0.012	0.044
	[0.667]	(0.086)	(0.043)	(0.044)	(0.043)
Life expectancy	81.438	-0.134	0.457	-0.119	-0.577
	[6.947]	(0.840)	(0.427)	(0.439)	(0.413)
Employed	0.898	-0.051	-0.041*	-0.047**	-0.006
	[0.303]	(0.043)	(0.022)	(0.022)	(0.022)
Income ($1000s)	6.866	-0.150	0.635	1.017	0.384
	[8.427]	(1.472)	(0.834)	(0.862)	(0.956)
$\mathbb{E}$[Wage in 5 years] /wage today	6.166	-0.636	-0.925	-1.422*	-0.491
	[11.685]	(1.761)	(0.710)	(0.734)	(0.667)
1(children)	0.774	0.033	-0.002	-0.014	-0.012
	[0.418]	(0.053)	(0.027)	(0.028)	(0.027)
Digit span recall	3.020	-0.156	-0.006	-0.010	-0.002
	[1.391]	(0.183)	(0.089)	(0.091)	(0.088)
Years of education	12.111	-0.468	0.161	-0.025	-0.186
	[2.869]	(0.371)	(0.184)	(0.189)	(0.181)
Primary school complete	0.964	-0.035	0.007	-0.022	-0.029**
	[0.186]	(0.026)	(0.013)	(0.013)	(0.013)
Secondary school complete	0.722	-0.047	0.024	-0.008	-0.031
	[0.449]	(0.058)	(0.029)	(0.029)	(0.028)
College degree	0.241	-0.054	-0.003	0.014	0.018
	[0.428]	(0.055)	(0.028)	(0.028)	(0.027)
N (exc. control in col. 2-4)	452	80	531	473	
Joint p-value		0.743	0.183	0.193	0.271

Standard deviations in brackets. Standard errors in parentheses. * $p < .10$, ** $p < .05$, *** $p < .01$

Column (1) reports the mean and standard deviation of the indicated variable across the control group. Columns (2) - (5) report the difference in mean of each variable across treatment arms, where terms in parentheses are standard errors of this pairwise t-test. The joint p-value reported at the bottom of this table reflects an F-test of joint orthogonality estimated by regressing treatment status on each of the covariates, imputing missing values with the median to avoid dropping the observations. The observation counts in columns 2-4 reflect the size of the treatment group compared to the control. 35 observations for which surveyors reported motorcycle taxi drivers pretending to be passengers are excluded.

Table 2: Effect of information on beliefs

	(1) Posterior risk	(2) Posterior risk	(3) Helmet effectiveness	(4) Helmet effectiveness	(5) Risk reduction	(6) Risk reduction, winsorized
Low treatment	-25.90	-22.19	-14.08	-13.87	-77.96	-43.14
	(21.77)	(21.51)	(0.97)	(0.96)	(18.31)	(13.84)
High treatment	6.77	1.04	-3.98	-3.41	-32.64	-2.95
	(24.91)	(23.88)	(0.88)	(0.88)	(19.08)	(15.51)
Control mean	330.97	330.97	78.68	78.68	270.44	228.62
Pr(High treatment = low treatment)	0.08	0.17	0.00	0.00	0.00	0.00
Observations	1,425	1,425	1,427	1,427	1,425	1,425
Controls	BL Risk	LASSO	BL Risk	LASSO	LASSO	LASSO
Enumerator FE	Yes	Yes	Yes	Yes	Yes	Yes

Robust Standard errors in parentheses.

Columns (1) - (2) report 10,000 times the respondent's stated posterior belief about the probability that they will suffer a fatal accident over the next 5 years, the lifespan of a helmet, if they do not purchase a helmet. Columns (3) - (4) report the respondent's stated belief about the effectiveness of motorcycle helmets, expressed as the percentage reduction in mortality risk. Columns (5) - (6) report 10,000 times the likelihood that a helmet will save the respondent's life over its 5 year lifespan based on their beliefs, the product of the prior two variables. The outcome in column (6) is winsorized at the 2nd and 98th percentiles. All estimates include wave fixed effects. Respondents assigned to the "Low treatment" group received the results of Liu et al. (2008) which estimates that helmets are 42% effective. Those assigned to "High treatment" were presented with the results of Ouellet and Kasantikul (2006) that helmets were 70% effective in a study in Thailand. Individuals in both treatment arms received empirical estimates of their risk of a fatal accident based on ridership. All models exclude 35 observations for which surveyors reported motorcycle taxi drivers pretending to be passengers.

Table 3: Value of a statistical life: Primary estimates

Panel A: Full sample

	(1) Interacted	(2) Interacted	(3) Treatment only	(4) Know victim
VSL	522.72	503.16	347.26	559.99
	(182.76)	(182.72)	(259.49)	(225.82)
Cragg-Donald F-stat	12.33	12.34	10.99	16.24
Weak IV Robust Confidence Set	[213.95, 1,042.75]	[189.98, 1,016.72]	[-160.17, 1,066.93]	NA
Confidence Set: Robust SEs	[10.05, 1,286.50]	[23.60, 1,309.24]	[-105.89, 1,003.85]	[193.40, 1,230.60]
Inversion test	CLR	CLR	CLR	AR
Observations	1,425	1,425	1,425	1,425
Controls	LASSO	BL Risk	BL Risk	None
Fixed effects	Enumerator	Enumerator	Enumerator	Terminal

Panel B: Estimates leaving 1 arm out

	(1) High treatment Interacted	(2) Low treatment Interacted	(3) Control Interacted	(4) Control Treatment only
VSL	383.39	576.99	787.15	721.11
	(179.71)	(273.64)	(542.92)	(453.95)
Cragg-Donald F-stat	17.00	9.92	3.89	10.46
Weak IV Robust Confidence Set	[42.70, 800.46]	[100.72, 1,405.38]	[-118.87, 5,160.10]	[-65.27, 2,225.04]
Inversion test	CLR	CLR	CLR	AR
Observations	963	905	982	982
Controls	BL Risk	BL Risk	BL Risk	BL Risk

Standard errors in parentheses.

Panel A reports regressions using observations in all experimental arms. Panel B reports results leaving one experimental arm out. "Interacted" columns use a first stage consisting of treatment and treatment interacted with baseline beliefs as instruments. Columns labeled "Treatment only" use only treatment assignment. All estimates control for baseline risk beliefs. Column (4) of panel A reports estimates that instrument for beliefs using exposure to a motorcycle accident victim, with robust Standard errors in parentheses. This instrument is speculative as knowing a victim may be endogenous, but is included to show that plausibly exogenous observational instruments align well with experimental estimates. Weak instrument robust confidence sets are reported in brackets, using the indicated tests. Weak IV and heteroskedastic robust confidence sets are also reported in panel A. Columns (1) - (3) of panel B include taxi terminal fixed effects. Excludes 35 observations where surveyors reported motorcycle taxi drivers pretending to be passengers.

Table 4: Heterogeneity in the value of a statistical life

	(1) 1(Age > median)	(2) asinh(Wage)	(3) 1(Wage > median)	(4) 1(health > median)	(5) 1(children)	(6) 1(digit recall score > median)	(7) 1(education > median)	(8) 1(female)
VSL	286.64	340.44	8.11	42.85	250.85	60.51	214.97	223.03
	(148.10)	(155.09)	(186.16)	(169.85)	(123.04)	(197.61)	(181.00)	(119.15)
VSL x Interaction	4.39	386.85	501.51	497.87	-97.05	297.54	70.90	260.45
	(268.81)	(127.27)	(273.24)	(254.27)	(247.69)	(242.17)	(232.93)	(238.62)
	[0.987]	[0.002]	[0.066]	[0.050]	[0.695]	[0.219]	[0.761]	[0.275]
VSL elasticity		1.14						
		(0.49)						
Cragg-Donald F-stat	17.72	12.37	18.78	17.01	21.70	12.10	18.78	16.78
Sanderson-Windmeijer first stage F-stats								
Base	36.18	19.58	33.87	48.22	31.56	16.14	25.07	27.87
Interaction	25.10	22.48	26.66	28.34	42.31	22.06	30.73	36.53
Observations	1,423	1,408	1,408	1,425	1,425	1,425	1,417	1,425
Controls	LASSO	LASSO	LASSO	LASSO	LASSO	LASSO	LASSO	LASSO

Standard errors in parentheses. P-values in brackets.

All columns report VSL estimates across the full sample and use a first stage consisting of treatment indicators and treatment indicators interacted with baseline motorcycle risk beliefs, and each of these values interacted with the demographic variable, as instruments for the mortality risk reduction of a helmet. The high treatment and control arms are pooled since posterior beliefs are similar among these groups, and once splitting on some dimensions of heterogeneity, the sample of respondents within certain groups otherwise becomes small since randomization was not stratified. In columns 1, 3-4, and 6-7, the demographic variable is converted to an indicator equal to 1 if the response was above the median. Unemployed individuals have wages coded to 0 in columns 2 and 3 since unemployment is typically involuntary in this sample. The estimates exclude 35 observations for which surveyors reported motorcycle taxi drivers pretending to be passengers.

Table 5: Correlates with beliefs

	(1) 10,000 × Risk	(2) 10,000 × Risk	(3) 10,000 × Risk	(4) 10,000 × Risk	(5) 10,000 × Risk
Panel A: No covariates					
Trips/week	-0.13				
	(3.71)				
Trip length		-1.01			
		(1.80)			
Previous accident			136.02		
			(47.02)		
Contact in accident				266.14	
				(37.21)	
Raining					301.83
					(172.92)
Sample average	333.22	333.22	333.22	333.22	333.22
Observations	1,427	1,427	1,427	1,427	1,427
Panel B: Taxi terminal FEs and controls					
Trips/week	-0.27				
	(3.98)				
Trip length		0.77			
		(1.65)			
Previous accident			90.05		
			(49.74)		
Contact in accident				204.19	
				(58.20)	
Raining					325.00
					(123.58)
Sample average	333.22	333.22	333.22	333.22	333.22
Observations	1,427	1,427	1,427	1,427	1,427

Robust Standard errors in parentheses. Standard errors in column 5 clustered by day.

Table 5 reports the correlation of demographic variables with prior beliefs about one's risk of dying in a motorcycle accident over a 5 year span. Estimates in Panel A do not include any controls or fixed effects. Estimates in Panel B include controls selected using double-post LASSO and taxi terminal/stand fixed effects (Belloni et al., 2014). Estimates interpolate missing controls using the median of the variable. Rainfall is coded to 1 if Visual Crossing reported positive precipitation from a weather station in Nairobi on the day the survey was conducted.

Table 6: Value of a statistical life: Estimates using alternative methods

Panel A: 42% helmet effectiveness treated as truth

	Rational expectations					Subjective beliefs
	(1) OLS	(2) LASSO	(3) LASSO	(4) LASSO	(5) $\frac{WTP_i}{\Delta r_i^*}$	(6) Beliefs
VSL	15,244.30	2,744.41	-6,167.09	1,490.08	381,326.62	110.96
	(10,900.69)	(10,630.65)	(10,840.79)	(10,747.15)	(13,859.27)	(53.74)
95% confidence set	[-6,121.06, 36,609.66]	[-18091.67, 23,580.49]	[-27415.04, 15,080.86]	[-19574.34, 22,554.50]	[354,162.44, 408,490.79]	[5.63, 216.29]
F-statistic						19.73
Observations	1,425	1,425	1,425	1,425	1,425	1,425
Controls	None	DPLASSO	DPLASSO	DPLASSO	None	None
Enumerator FE	No	No	Yes	No	No	Yes
Taxi Terminal FE	No	No	No	Yes	No	No

Panel B: 70% helmet effectiveness treated as truth

	Rational expectations					Subjective beliefs
	(1) OLS	(2) LASSO	(3) LASSO	(4) LASSO	(5) $\frac{WTP_i}{\Delta r_i^*}$	(6) Beliefs
VSL	9,146.58	1,646.65	-3,700.25	894.05	228,795.97	110.96
	(6,540.42)	(6,378.39)	(6,504.48)	(6,448.29)	(8,315.56)	(53.74)
95% confidence set	[-3,672.63, 21,965.79]	[-10855.00, 14,148.30]	[-16449.03, 9,048.52]	[-11744.61, 13,532.70]	[212,497.47, 245,094.48]	[5.63, 216.29]
F-statistic						19.73
Observations	1,425	1,425	1,425	1,425	1,425	1,425
Controls	None	DPLASSO	DPLASSO	DPLASSO	None	None
Enumerator FE	No	No	Yes	No	No	Yes
Taxi Terminal FE	No	No	No	Yes	No	No

Robust Standard errors in parentheses.

Columns (1) - (4) report VSL estimates obtained using equation 11. Column (5) estimates VSL using equation 12, reporting the mean. Panel A uses empirical helmet effectiveness from Liu et al. (2008). Panel B uses Ouellet and Kasantikul (2006). Column (6) reports VSL estimates using beliefs about deaths/10,000 riders to instrument for agents' beliefs about the risk reduction of a helmet. 35 observations where motorcycle taxi drivers pretended to be passengers are excluded.

Online Appendix

A Belief elicitation

The survey consisted of five modules. First, we collected demographic information. Second, we obtained information about use of motorcycle taxis, including ridership volume, trip length and types, and reasons for using motorcycle taxis. Third, surveyors elicited the respondents' beliefs about their likelihood of dying in a motorcycle taxi accident. The first part of this module was identical across the treatment and control arms. I refer to variables measured during this portion as baseline beliefs. During the second component, surveyors presented empirical risk estimates to respondents in the treatment arms and elicited posterior beliefs. Fourth, surveyors presented individuals in the treatment arms with information about helmet efficacy and then measured posterior beliefs about the effectiveness of helmets across the control and treatment arms. Finally, respondents completed the BDM game and received a helmet or mobile money payment. The third and fourth modules were skipped for those in the pure control group.

Prior to the first survey wave, surveyors completed a one week pilot that was focused on identifying a reliable survey module to measure beliefs about mortality risk. The final set of questions begins by providing the passengers with reference points to help them express and contextualize rare events. We informed the respondents that Jamhuri (Independence) Day occurs one out of every 365.25 days and that a leap day occurs one out of 1,461 days.

Surveyors next asked respondents how many fatal accidents occur per 100,000 taxi trips. In addition to measuring the respondent's views about per trip risks, this question was designed to help the passengers think carefully about mortality risks by walking them through first thinking about how dangerous each trip is, then about their volume of trips. We then asked the respondent how many deaths there are per 10,000 passengers over the course of 1 year and 5 years in Nairobi.

We next asked the respondents about their own risk of suffering a fatal accident over the following 5 years. We did this in two steps. First, we asked them to select which range of risks seemed most accurate, for instance less than 1 in 10,000,000, between 1 in 10,000,000 and 1 in 1,000,000, etc. After selecting a range, we asked the passengers to respond with their belief within the range. Piloting revealed that this two step approach helped respondents answer accurately.

The respondents were then asked which information sources they used to construct their beliefs

and whether they had been in a previous accident. We then presented those in the treatment groups with empirical estimates of their 5-year fatal accident risk as a function of their ridership, then elicited posterior beliefs. Empirical estimates were constructed by normalizing the number of recorded deaths by the NTSA the year before the study across Kenya by the total estimated number of motorcycle trips to get a per trip risk estimate. Then we used the respondent's reports of their ridership to estimate risk over the 5 year helmet lifespan.

The motorcycle taxi context was chosen in part because empirical risks are high enough over the lifespan of a helmet to limit problems associated with understanding small probabilities. For a passenger that takes 6 trips per week, the median in this sample, we estimate that there is over a 1 in 5,000 chance that they will die in a motorcycle taxi accident in the next 5 years absent a helmet. Given limitations of the data used to construct this estimate, this may be a lower bound on the true risk. While this is still a relatively low probability, it is among the most probable causes of death for young adults and thus minimizes the cognitive burden of understanding small probabilities compared to other settings.

The efficacy of helmets is easier to comprehend, so we followed a simpler procedure. We first presented the low treatment with the Liu et al. (2008) estimate that helmets reduce one's likelihood of dying by 42% and the high treatment with the Ouellet and Kasantikul (2006) result that helmets reduce mortality risk by 70%. The control and treatment groups were then asked for their own beliefs about how effectively helmets prevent death, expressed as the number of people that they believe would survive if all passengers wore a helmet per 100 deaths if no one wore a helmet.

B Local average VSLs identified by interacted and non-interacted instruments

This section derives the weighted average VSLs identified by the treatment only and interacted vectors of instruments. I begin with the case where $r_{0,i}$ and Δr_i are measured without systematic error. Since agents do not update beliefs about the unhelmeted risk of motorcycles when presented with information, their posterior beliefs about the likelihood that a helmet will save their life is approximately given by

$$\Delta r_i = HE_i(T_i) r_{0,i} + \nu_i$$

where $r_{0,i}$ is their prior about the mortality risk of motorcycles and HE_i is the posterior belief about the efficacy of a helmet. Letting L_i indicate assignment to the low treatment arm and H_i to

the high treatment arm, the relationship is therefore approximated by

$$\Delta r_i = \pi_c r_{0,i} + \pi_L L_i \cdot r_{0,i} + \pi_H H_i \cdot r_{0,i} + \nu_i$$

One may also estimate a treatment only first stage

$$\Delta r_i = \pi_1 L_i + \pi_2 H_i + \nu_i'$$

And an individual's helmet valuation can be written as

$$v_i = \alpha + VSL_i \Delta r_i + \epsilon_i$$

I apply the Frisch-Waugh-Lovell theorem to partial out the constant in the treatment only first stage and $r_{0,i}$ from the interacted, denoting by $\tilde{x}_i$ the partialled out variables.

Beginning with the treatment only IV, substitution of the known form for Δr_i into the structural equation for v_i yields

$$\begin{aligned}
\hat{VSL}^{TO}_{2SLS} &= \frac{Cov\left(v_i, \pi_1 \tilde{L}_i + \pi_2 \tilde{H}_i\right)}{Cov\left(\Delta r_i, \pi_1 \tilde{L}_i + \pi_2 \tilde{H}_i\right)} \\
&= \frac{Cov\left(VSL_i \pi_L L_i \cdot r_{0,i} + VSL_i \pi_H H_i \cdot r_{0,i}, \pi_1 \tilde{L}_i + \pi_2 \tilde{H}_i\right)}{\mathbb{E}\left[r_{0,i}\right]\left(\pi_L \pi_1 V(L_i) + \pi_H \pi_2 V(H_i) + (\pi_L \pi_2 + \pi_H \pi_1) Cov(L_i, H_i)\right)} \\
&= \mathbb{E}\left[VSL_i \frac{r_{0,i}\left(\pi_1 \pi_L V(L_i) + \pi_2 \pi_H V(H_i) + (\pi_L \pi_2 + \pi_H \pi_1) Cov(L_i, H_i)\right)}{\mathbb{E}\left[r_{0,i}\right]\left(\pi_1 \pi_L V(L_i) + \pi_2 \pi_H V(H_i) + (\pi_L \pi_2 + \pi_H \pi_1) Cov(L_i, H_i)\right)}\right] \\
&= \mathbb{E}\left[VSL_i \frac{r_{0,i}}{\mathbb{E}\left[r_{0,i}\right]}\right]
\end{aligned}$$

Showing individuals are weighted linearly in priors, $r_{0,i}$. In the interacted case,

$$\begin{aligned}
\hat{VSL}^{INT}_{2SLS} &= \frac{Cov\left(v_i, \pi_3 (L_i \tilde{\cdot} r_{0,i}) + \pi_4 (H_i \tilde{\cdot} r_{0,i})\right)}{Cov\left(\Delta r_i, \pi_3 (L_i \tilde{\cdot} r_{0,i}) + \pi_4 (H_i \tilde{\cdot} r_{0,i})\right)} \\
&= \frac{Cov\left(VSL_i \pi_L L_i \cdot r_{0,i} + VSL_i \pi_H H_i \cdot r_{0,i}, \pi_3 (L_i \tilde{\cdot} r_{0,i}) + \pi_4 (H_i \tilde{\cdot} r_{0,i})\right)}{\mathbb{E}\left[r_{0,i}^2\right]\left(\pi_L \pi_3 V(L_i) + \pi_H \pi_4 V(H_i) + (\pi_L \pi_4 + \pi_H \pi_3) Cov(L_i, H_i)\right)} \\
&= \mathbb{E}\left[VSL_i \frac{r_{0,i}^2\left(\pi_3 \pi_L V(L_i) + \pi_4 \pi_H V(H_i) + (\pi_L \pi_4 + \pi_H \pi_3) Cov(L_i, H_i)\right)}{\mathbb{E}\left[r_{0,i}^2\right]\left(\pi_3 \pi_L V(L_i) + \pi_4 \pi_H V(H_i) + (\pi_L \pi_4 + \pi_H \pi_3) Cov(L_i, H_i)\right)}\right] \\
&= \mathbb{E}\left[VSL_i \frac{r_{0,i}^2}{\mathbb{E}\left[r_{0,i}^2\right]}\right]
\end{aligned}$$

The interacted instruments weight observations proportionally to priors squared, $r_{0,i}^2$. The primary VSL estimates use a measure of $r_{0,i}$ distinct from that in Δr_i to avoid attenuation from measurement error passed through to the instrument. This may lead to a LATE that weights observations between stylized limit cases, but the instrument still weights observations more heavily in priors so the logic persists. The test results hold using the same measure of $r_{0,i}$ used to construct Δr_i.

VSL heterogeneity correlated with $r_{0,i}$ could confound this test in certain cases. However, to lead the test to fail to reject, VSL heterogeneity would need to offset misreporting, which is unlikely. The primary use of this test is also to test for large misreporting that could cause the VSL estimates to fall well below those used in policy when in fact the true value is orders of magnitude larger. While issues like heterogeneity in VSL may make the test less effective at isolating smaller bias, they are unlikely to lead it to fail to reject if belief mismeasurement caused orders of magnitude bias in estimates.

Misreported mortality risk:

Suppose that agents report $r_{0,i}\zeta_i$ for some $\zeta_i > 0$ due to systematic measurement error but the true beliefs guiding valuations are $r_{0,i}$. This captures challenges reporting small probabilities. One can show that

$$\hat{VSL}_{2SLS}^{TO} = \mathbb{E}\left[VSL_i \frac{r_{0,i}}{\mathbb{E}[r_{0,i}\zeta_i]}\right]$$
$$\hat{VSL}_{2SLS}^{INT} = \mathbb{E}\left[VSL_i \frac{r_{0,i}^2\zeta_i}{\mathbb{E}[r_{0,i}^2\zeta_i^2]}\right]$$

Therefore one typically has $\hat{VSL}_{2SLS}^{TO} \neq \hat{VSL}_{2SLS}^{INT}$ unless $\zeta_i = c$. If individuals with low beliefs round down and those with high beliefs round up, $Cov(\zeta_i, r_{0,i}) > 0$. If agents always round small probabilities up or down, $Cov(\zeta_i, r_{0,i}) \neq 0$.

C Classification of NBER Working Papers Applying VSL

This appendix details the steps taken to classify NBER working papers that apply VSL. This exercise was used to assess whether VSL is typically applied to theoretically supported decisions or not when used in low and middle-income countries. The classification was done with Claude and may not perfectly replicate given randomization injected into large language models. This appendix was written in part by Claude, which I tasked with detailing the steps it took carefully to allow

others to replicate the analysis as closely as possible.

C.1 Paper selection

I queried every NBER paper since 2000 whose text contained the term "value of a statistical life," "VSL", "disability-adjusted life year," or "DALY." The search was run using EconLit in August 2026. The search returned 363 papers. Each was converted from PDF to Markdown (using `marker-pdf`), then the markdown was run through the classification procedure below.

C.2 Two-stage classification

Stage 1 (screening). Every candidate paper was coded on two binary variables:

- `estimate_vsl`: the paper estimates VSL
- `use_vsl`: the paper applies VSL e.g. in a benefit-cost analysis, to evaluate a policy, or to compute a welfare effect. Computing a cost-per-DALY or DALYs-averted figure counts as an application; a literature-review mention with no application does not.

Papers coded `use_vsl` = 1 (240 of 363) proceeded to Stage 2.

Stage 2 (misapplication classification). Each `use_vsl` = 1 paper was further coded on:

- `developing_country`: VSL/DALYs are applied to a low- or middle-income population, or to a global/cross-country policy that includes such populations.
- `nonmarginal_risk`: VSL is applied to a decision that is not a marginal (small) change in mortality risk – e.g. a curative treatment, a large discrete intervention, or aggregation across a catastrophic event.
- `transfer`: the paper prices a population's lives using that population's own (possibly income-adjusted) VSL, where that valuation feeds a decision actually financed or decided by a different, wealthier entity – e.g. a rich country, a multilateral donor, or a globally coordinated process dominated by rich countries.
- `transfer_correct`: the paper applies a funder's VSL to value benefits in a population it is externally financing.
- `self_aware`: relevant only when `nonmarginal_risk` or `transfer` equals 1; the authors explicitly flag the application as imperfect.
- `givewell`: the study evaluates an intervention financed, in whole or part, by GiveWell.

For papers with $developing_country = 1$, I also instructed Claude to code $transfer_category$, classifying the paper's subject matter as *health_program*, *pollution* (air or water pollution, emissions, or climate change), or *other*.

C.3 Replication: steps and prompts

The exercise was performed using Claude Sonnet 5. Each paper was classified by a separate Claude Code subagent, so that each classification decision was made independently and without contamination from other papers' context. In outline:

1. **Run Stage 1.** For each paper, launch one Claude Code `Agent` subagent running Claude Sonnet 5 with the Stage 1 prompt template below, in parallel batches of up to 25 papers.
2. **Aggregate.** Collect the returned lines into a running Stage 1 results table.
3. **Filter and run Stage 2.** Restrict to papers coded `use_vsl` = 1 not yet in the Stage 2 table, and repeat classification steps with the Stage 2 prompt template, again one Claude Sonnet 5 subagent per paper.
4. **Spot-check.** Periodically re-verify a sample of negative classifications by hand; if a systematic tooling issue is found, re-audit the affected batches.
5. **Manual verification.** I checked the 21 `transfer` = 1 cases. w34357 was determined to be a review never applying VSL. The remainder either used VSL (or DALY) directly or referenced cost-effectiveness thresholds in terms of cost per DALY averted, a conceptually similar practice.

Stage 1 prompt.

```
Classify this paper for Stage 1, using the Stage 1 definitions of estimate_vsl
and use_vsl.

Paper: <paper ID> ("<title>") File: <path to converted markdown>

Search the full text for "statistical life", "VSL", "DALY" and
"disability-adjusted life" first, then read more broadly if needed.
Use a search tool that does not skip files silently.

Respond with EXACTLY ONE line: <DOI>,<estimate_vsl 0/1>,<use_vsl 0/1>
Then a REASON line quoting the textual evidence.
```

Stage 2 prompt.

```
Classify this paper for Stage 2, using the Stage 2 definitions. The
paper is already coded use_vsl=1.

Paper: <paper ID> ("<title>") File: <path to converted markdown>

Code six binary columns: developing_country, nonmarginal_risk, transfer,
transfer_correct, self_aware, givewell. If developing_country=1,
also code transfer_category.

Calibrate `transfer` against these anchors:
  - w27599: region's own VSL for that region's own adaptation
    spending = 0; the same VSLs aggregated into a global social cost
    of carbon benchmarked against US regulatory policy = 1. Code 1 if
    ANY substantive application qualifies, and say the case is mixed.
  - w28734: GiveWell-funded Bangladesh trial benchmarked against
    Bangladesh's own $205,000 VSL = 1.
  - w34081: global income-scaled VSL aggregate used only as
    introductory motivation, never entering the model or a policy
    comparison = 0.
  - w24247: descriptive cross-country DALY panel attached to no
    financing or policy decision = 0.
  - Any entity valuing only its own population, which it both decides
    for and finances = 0, however income-adjusted.

Search the full text for "statistical life", "VSL", "DALY", "fund",
"financ" and "polic" with surrounding context first, then read more
broadly if needed. Use a search tool that does not skip files
silently.

Respond with EXACTLY ONE line:
<DOI>,<developing_country>,<nonmarginal_risk>,<transfer>,
<transfer_correct>,<self_aware>,<givewell>[,<transfer_category>]
Then brief REASON lines quoting textual evidence for each column.
```

C.4 Full classification results

Paper	Dev. Country	Non-marg. Risk	Transfer	Transfer Correct	Self-Aware	GiveWell	Category
w7716	Y	–	Y	–	–	–	Health
w7717	Y	–	Y	–	–	–	Health
w7760	–	Y	–	–	Y	–	–
w8481	Y	Y	Y	–	–	–	Health
w8755	–	–	–	–	–	–	–
w8818	–	Y	–	–	–	–	–
w9396	–	Y	–	–	–	–	–
w9487	–	–	–	–	–	–	–
w9765	Y	Y	–	–	–	–	Health
w10053	–	–	–	–	–	–	–
w10328	–	Y	–	–	–	–	–
w10511	Y	Y	–	–	–	–	Health
w10737	–	Y	–	–	–	–	–
w10824	–	–	–	–	–	–	–
w10906	–	–	–	–	–	–	–
w11288	Y	–	Y	–	–	–	Health
w11316	–	–	–	–	–	–	–
w11405	–	Y	–	–	–	–	–
w11934	–	–	–	–	–	–	–
w11956	–	–	–	–	–	–	–
w12054	–	Y	–	–	Y	–	–
w12092	–	Y	–	–	–	–	–
w12122	Y	Y	–	–	–	–	Other
w12184	–	Y	–	–	–	–	–
w12294	–	–	–	–	–	–	–
w12836	–	Y	–	–	Y	–	–
w13087	Y	Y	–	–	–	–	Health
w13178	–	–	–	–	–	–	–
w13396	Y	Y	Y	–	Y	–	Health
w13490	–	Y	–	–	Y	–	–
w13599	–	Y	–	–	Y	–	–
w13877	–	–	–	–	–	–	–
w14011	Y	Y	–	–	–	–	Pollution
w14093	Y	Y	–	–	Y	–	Health
w14504	–	–	–	–	–	–	–
w14522	–	Y	–	–	–	–	–
w14823	–	Y	–	–	–	–	–
w14977	–	–	–	–	–	–	–
w15174	–	Y	–	–	–	–	–
w15235	–	Y	–	–	–	–	–
w15280	Y	–	–	–	–	–	Health
w15383	–	–	–	–	–	–	–
w15471	–	Y	–	–	Y	–	–
w15574	–	Y	–	–	–	–	–
w15649	–	Y	–	–	Y	–	–
w15658	–	–	–	–	–	–	–
w15666	–	–	–	–	–	–	–
w16108	–	Y	–	–	–	–	–
w16352	Y	Y	–	–	Y	–	Other
w16599	Y	Y	–	–	Y	–	Other
w16722	–	–	–	–	–	–	–
w16844	–	–	–	–	–	–	–
w16953	–	Y	–	–	–	–	–
w17094	–	Y	–	–	–	–	–
w17170	–	–	–	–	–	–	–
w17937	–	Y	–	–	–	–	–
w18267	–	–	–	–	–	–	–
w18468	–	Y	–	–	Y	–	–
w18815	–	–	–	–	–	–	–
w18849	–	–	–	–	–	–	–

Paper	Dev. Country	Non-marg. Risk	Transfer	Transfer Correct	Self-Aware	GiveWell	Category
w19058	–	–	–	–	–	–	–
w19494	Y	–	–	–	–	–	Other
w19881	–	–	–	–	–	–	–
w19956	–	Y	–	–	–	–	–
w20215	–	Y	–	–	Y	–	–
w20330	–	Y	–	–	–	–	–
w20460	Y	–	–	–	–	–	Pollution
w20500	–	–	–	–	–	–	–
w20525	–	–	–	–	–	–	–
w20653	–	Y	–	–	–	–	–
w20810	–	–	–	–	–	–	–
w21129	–	–	–	–	–	–	–
w21291	–	–	–	–	–	–	–
w21308	–	Y	–	–	Y	–	–
w21324	Y	Y	Y	–	–	–	Health
w21383	–	–	–	–	–	–	–
w21635	–	Y	–	–	–	–	–
w21671	–	–	–	–	–	–	–
w22092	–	–	–	–	–	–	–
w22105	Y	–	–	–	–	–	Pollution
w22155	–	–	–	–	–	–	–
w22261	–	–	–	–	–	–	–
w22579	–	–	–	–	–	–	–
w22610	–	–	–	–	–	–	–
w22769	–	–	–	–	–	–	–
w22796	–	–	–	–	–	–	–
w22814	–	–	–	–	–	–	–
w22899	–	–	–	–	–	–	–
w23340	–	–	–	–	–	–	–
w23346	–	Y	–	–	Y	–	–
w23386	–	–	–	–	–	–	–
w23413	–	–	–	–	–	–	–
w23417	–	–	–	–	–	–	–
w23791	–	–	–	–	–	–	–
w23852	–	–	–	–	–	–	–
w23866	–	–	–	–	–	–	–
w23873	–	–	–	–	–	–	–
w23966	–	–	–	–	–	–	–
w24183	Y	–	–	–	–	–	Pollution
w24206	Y	Y	Y	–	–	–	Health
w24247	Y	Y	–	–	Y	–	Health
w24308	–	–	–	–	–	–	–
w24607	–	Y	–	–	–	–	–
w24688	Y	–	–	–	–	–	Pollution
w24802	–	Y	–	–	–	–	–
w25055	–	Y	–	–	–	–	–
w25241	–	–	–	–	–	–	–
w25339	–	–	–	–	–	–	–
w25641	–	–	–	–	–	–	–
w25661	–	–	–	–	–	–	–
w25731	Y	–	Y	–	–	–	Health
w25791	–	–	–	–	–	–	–
w25910	–	–	–	–	–	–	–
w25971	–	Y	–	–	–	–	–
w26068	–	Y	–	–	Y	–	–
w26381	–	–	–	–	–	–	–
w26775	Y	Y	Y	–	–	–	Health
w26783	–	–	–	–	–	–	–
w26804	–	–	–	–	–	–	–
w26882	–	Y	–	–	–	–	–

Paper	Dev. Country	Non-marg. Risk	Transfer	Transfer Correct	Self-Aware	GiveWell	Category
w26942	–	Y	–	–	–	–	–
w26981	–	Y	–	–	Y	–	–
w26984	–	Y	–	–	–	–	–
w27009	–	Y	–	–	–	–	–
w27046	–	Y	–	–	–	–	–
w27059	–	Y	–	–	–	–	–
w27062	–	Y	–	–	–	–	–
w27104	–	Y	–	–	–	–	–
w27121	–	Y	–	–	Y	–	–
w27135	–	–	–	–	–	–	–
w27172	–	–	–	–	–	–	–
w27245	–	–	–	–	–	–	–
w27285	–	–	–	–	–	–	–
w27340	–	Y	–	–	–	–	–
w27398	Y	–	–	–	–	–	Pollution
w27430	–	–	–	–	–	–	–
w27458	–	Y	–	–	–	–	–
w27578	–	–	–	–	–	–	–
w27599	Y	–	Y	–	–	–	Pollution
w27757	Y	Y	–	–	Y	–	Health
w27794	–	Y	–	–	–	–	–
w27848	–	Y	–	–	–	–	–
w27863	–	Y	–	–	Y	–	–
w27933	–	–	–	–	–	–	–
w28031	–	Y	–	–	–	–	–
w28064	–	Y	–	–	–	–	–
w28071	–	–	–	–	–	–	–
w28199	–	–	–	–	–	–	–
w28202	–	–	–	–	–	–	–
w28282	–	Y	–	–	–	–	–
w28492	Y	Y	Y	–	–	–	Health
w28550	–	–	–	–	–	–	–
w28601	–	Y	–	–	Y	–	–
w28617	–	–	–	–	–	–	–
w28619	–	–	–	–	–	–	–
w28622	–	–	–	–	–	–	–
w28636	–	Y	–	–	Y	–	–
w28707	–	–	–	–	–	–	–
w28708	Y	–	–	–	–	–	Pollution
w28734	Y	–	Y	–	–	Y	Health
w28848	–	–	–	–	–	–	–
w28867	–	Y	–	–	–	–	–
w29063	–	Y	–	–	–	–	–
w29071	–	–	–	–	–	–	–
w29104	–	Y	–	–	–	–	–
w29152	–	–	–	–	–	–	–
w29419	–	–	–	–	–	–	–
w29454	–	–	–	–	–	–	–
w29465	–	–	–	–	–	–	–
w29574	–	Y	–	–	–	–	–
w29682	Y	Y	–	–	–	–	Health
w29718	–	Y	–	–	Y	–	–
w29764	–	Y	–	–	Y	–	–
w29809	–	Y	–	–	–	–	–
w29845	–	–	–	–	–	–	–
w29854	–	–	–	–	–	–	–
w29884	–	–	–	–	–	–	–
w30104	–	Y	–	–	–	–	–
w30118	–	–	–	–	–	–	–
w30181	–	–	–	–	–	–	–

Paper	Dev. Country	Non-marg. Risk	Transfer	Transfer Correct	Self-Aware	GiveWell	Category
w30286	–	–	–	–	–	–	–
w30303	–	Y	–	–	–	–	–
w30391	Y	–	Y	–	–	–	Health
w30553	–	–	–	–	–	–	–
w30565	Y	Y	–	–	–	–	Health
w30619	Y	Y	Y	–	–	–	Health
w30626	–	–	–	–	–	–	–
w30648	Y	–	–	–	–	–	Pollution
w30663	–	Y	–	–	–	–	–
w30670	–	–	–	–	–	–	–
w30702	–	–	–	–	–	–	–
w30801	–	–	–	–	–	–	–
w30830	–	–	–	–	–	–	–
w30852	–	Y	–	Y	Y	–	–
w30966	–	–	–	–	–	–	–
w31139	–	–	–	–	–	–	–
w31162	Y	–	Y	–	–	Y	Health
w31203	Y	–	–	–	–	–	Health
w31361	–	–	–	–	–	–	–
w31379	–	–	–	–	–	–	–
w31423	Y	Y	–	–	–	–	Health
w31475	–	–	–	–	–	–	–
w31555	–	Y	–	–	–	–	–
w31593	–	–	–	–	–	–	–
w31758	–	Y	–	–	Y	–	–
w31809	Y	–	–	–	–	–	Pollution
w31812	Y	Y	Y	–	Y	–	Health
w31885	–	–	–	–	–	–	–
w31957	–	Y	–	–	–	–	–
w31999	Y	Y	–	–	Y	–	Other
w32035	–	–	–	–	–	–	–
w32059	–	Y	–	–	–	–	–
w32110	–	–	–	–	–	–	–
w32195	–	–	–	–	–	–	–
w32235	–	–	–	–	–	–	–
w32349	–	Y	–	–	Y	–	–
w32548	–	–	–	–	–	–	–
w32558	Y	Y	–	–	–	–	Health
w32613	–	–	–	–	–	–	–
w32728	Y	–	Y	–	–	–	Pollution
w32742	Y	Y	Y	–	–	–	Other
w32831	–	Y	–	–	–	–	–
w32870	–	Y	–	–	–	–	–
w32955	–	–	–	–	–	–	–
w32984	Y	Y	–	–	–	–	Health
w33032	–	–	–	–	–	–	–
w33143	Y	–	–	–	–	–	Pollution
w33249	–	Y	–	–	–	–	–
w33279	–	Y	–	–	–	–	–
w33429	–	Y	–	–	Y	–	–
w33447	–	–	–	–	–	–	–
w33557	Y	–	–	–	–	–	Health
w33602	–	Y	–	–	–	–	–
w33699	–	Y	–	–	–	–	–
w33905	–	Y	–	–	–	–	–
w34081	Y	Y	–	–	Y	–	Pollution
w34152	Y	Y	Y	–	Y	Y	Other
w34339	Y	–	Y	–	–	–	Other
w34357	Y	–	Y	–	–	–	Pollution
w34525	–	–	–	–	–	–	–

Notes: Papers labeled by NBER working paper number. This table displays the raw classification outputs produced by the language model. As discussed in the text, paper w34357 was manually identified as a false positive (a review rather than an application) and is therefore excluded from the final prevalence calculation, leaving 20 transfer cases out of 50 total applications.

D Welfare losses using recipient versus donor VSL

Consider a population of consumers that share the same preferences, but differ in income levels. An individual with consumption level c has marginal utility of consumption $\alpha(c)$, and their utility gain from a marginal reduction in mortality risk is $\beta(c) = u_a(c) - u_d$. The parameters have constant elasticity relationships with respect to consumption, $\frac{d\log\alpha(c)}{d\log c} = \eta_{\alpha,c}$ and $\frac{d\log\beta(c)}{d\log c} = \eta_{\beta,c}$.

Individuals are located in a rich country with consumption level c_h or a poor country with consumption level $c_\ell < c_h$. The goal is to calculate the welfare gains of a health intervention implemented in ℓ but paid for by consumers in h. Denote $R = \frac{c_h}{c_\ell}$, $\alpha(c_i) = \alpha_i$ and $\beta(c_i) = \beta_i$.

The program costs ζ and saves ρ lives. $VSL_h = \frac{\beta_h}{\alpha_h}$ and $VSL_\ell = \frac{\beta_\ell}{\alpha_\ell}$ are observed. $VSL^* = \frac{\beta_\ell}{\alpha_h}$ is not. The oracle benefit-cost ratio of the program is given by $r^* = \frac{\rho VSL^*}{\zeta}$. Observe that this value exceeds 1 if and only if $\beta_\ell\rho > \alpha_h\zeta$, that is, if the utility gain to recipients exceeds the utility cost to donors.

A utilitarian social planner, weighting donors and recipients equally, maximizes welfare, implementing a program if and only if its benefit-cost ratio $r \geq 1$, where $r = \frac{\rho VSL}{\zeta}$ for whichever VSL the analyst uses. The infeasible oracle rule would implement a program if and only if $r^* \geq 1$. The welfare loss, priced in donor consumption, from using VSL in place of VSL^* is

$$L(VSL) \;=\; \begin{cases} \zeta\,|r^* - 1| & \text{if the rule based on } VSL \text{ differs from the rule based on } VSL^*, \\ 0 & \text{otherwise.} \end{cases}$$

Proposition 1 *Suppose $R \geq 2$, $\eta_{\beta,c} > 0$, and $\eta_{\alpha,c} \leq -1$. Then*

$$\sup_{r^*}\; L(VSL_h) \;=\; \zeta\left(1 - R^{-\eta_{\beta,c}}\right) \;<\; \zeta \;\leq\; \zeta\left(R^{|\eta_{\alpha,c}|} - 1\right) \;=\; \sup_{r^*}\; L(VSL_\ell),$$

so VSL_h minimax dominates VSL_ℓ.

Proof. Constant elasticities give

$$\frac{VSL^*}{VSL_\ell} = R^{|\eta_{\alpha,c}|} \qquad \text{and} \qquad \frac{VSL_h}{VSL^*} = R^{\eta_{\beta,c}}.$$

Consequently the rule based on VSL_h funds if and only if $r^* > R^{-\eta_{\beta,c}}$.

Since $\eta_{\beta,c} > 0$ and $R > 1$ we have $0 < R^{-\eta_{\beta,c}} < 1$, differing from the oracle on $r^* \in$

$(R^{-\eta_{\beta,c}}, 1)$. This is strictly decreasing in r^*, so

$$\sup_{r^*} L(VSL_h) \;=\; \lim_{r^* \searrow R^{-\eta_{\beta,c}}} \zeta(1 - r^*) \;=\; \zeta\left(1 - R^{-\eta_{\beta,c}}\right) \;<\; \zeta.$$

The rule based on VSL_ℓ funds iff $r^* > R^{|\eta_{\alpha,c}|} > 1$, differing from the oracle on $r^* \in (1, R^{|\eta_{\alpha,c}|})$. This is strictly increasing in r^*, so

$$\sup_{r^*} L(VSL_\ell) \;=\; \lim_{r^* \nearrow R^{|\eta_{\alpha,c}|}} \zeta(r^* - 1) \;=\; \zeta\left(R^{|\eta_{\alpha,c}|} - 1\right).$$

Since $R \geq 2$ and $|\eta_{\alpha,c}| \geq 1$, $R^{|\eta_{\alpha,c}|} \geq 2$ so $\zeta(R^{|\eta_{\alpha,c}|} - 1) \geq \zeta$. □

Now suppose that the true benefit-cost ratios of programs have density f independent of costs. Set $B = R^{-\eta_{\beta,c}} \in (0, 1)$ and $A = R^{|\eta_{\alpha,c}|} \geq 2$.

Proposition 2 *Suppose $R \geq 2$, $\eta_{\beta,c} > 0$, and $\eta_{\alpha,c} \leq -1$. Then $\mathbb{E}[L(VSL_h)] \leq \mathbb{E}[L(VSL_\ell)]$ if and only if*

$$\underbrace{\int_B^1 (1 - r^*) f(r^*)\, dr^*}_{\textit{waste on programs funded in error}} \quad \leq \quad \underbrace{\int_1^A (r^* - 1) f(r^*)\, dr^*}_{\textit{surplus forgone on programs rejected in error}} .$$

Proof. From above, the rule based on VSL_h differs from the oracle iff $r^* \in (B, 1)$, with loss $\zeta(1 - r^*)$, and the rule based on VSL_ℓ differs iff $r^* \in (1, A)$, with loss $\zeta(r^* - 1)$. Hence

$$\mathbb{E}[L(VSL_h)] = \zeta \int_B^1 (1 - r^*) f(r^*)\, dr^*, \qquad \mathbb{E}[L(VSL_\ell)] = \zeta \int_1^A (r^* - 1) f(r^*)\, dr^*,$$

yielding the stated condition. □

Corollary 1 *If f is symmetric about 1, then $\mathbb{E}[L(VSL_\ell)] \geq \mathbb{E}[L(VSL_h)]$.*

Proof. Substituting $r^* = 1 - x$ on $(B, 1)$ and $r^* = 1 + x$ on $(1, 2 - B)$, both with x over $(0, 1 - B)$, symmetry gives $\int_B^1 (1 - r^*) f(r^*)\, dr^* = \int_0^{1-B} x f(1 + x)\, dx = \int_1^{2-B} (r^* - 1) f(r^*)\, dr^*$, which is at most $\int_1^A (r^* - 1) f(r^*)\, dr^*$ since the integrand is non-negative on $(2 - B, A)$. □

E Ethics Appendix

This appendix describes ethical considerations taken to prevent harming respondents.

The helmets used in the study were manufactured by Boda Plus, a local Kenyan producer. I was referred to this brand by a road safety NGO, which believed that they were high quality. I verified

that the helmets adhered to the Kenyan Bureau of Standards safety regulations before the study. Independent journalists have also concluded that Boda Plus manufactures high quality helmets. For instance, Weronika Strzyzynska reports in *The Guardian* article "Africa sees sharp rise in road traffic deaths as motorbike taxis boom" (2023) that "Kenyan company Boda Plus's helmets meet high safety standards but can't compete on price with low-quality imported ones."

The first result presented to respondents, that helmets reduce mortality risk by 42% as reported in Liu et al. (2008), was selected because it is often presented to consumers by the Kenyan Government, NGOs and the UN Road Safety Fund. To give one example, the FIA Foundation cites this figure in their post "Helmets testing and awareness needed to curb Kenyan motorcycle deaths, says report supported by FIA Foundation."

The second result presented to respondents is from Ouellet and Kasantikul (2006) which finds that helmeted drivers have about a 70% lower mortality rate in Thailand. Although Liu et al. (2008) is often cited, the underlying studies used in the meta analysis are primarily from high-income settings. In such cases, motorcycles drive faster and helmets are higher quality, so there may be different effectiveness versus LMICs. I selected Ouellet and Kasantikul (2006) based on this logic, which was the most relevant LMIC that I could find. I also considered the studies from developing countries analyzed in Liu et al. (2008), but they are older or do not report mortality rates. Liu et al. (2008) did not identify any RCT evidence about helmet efficacy, possibly due to ethical concerns with a control group, which is why randomized evidence is not used. Liu et al. (2008) main inclusion criteria is that studies have a clear control group: the authors note that most papers fail to account for selection, and a large share use data only on those that end up hospitalized, which fails to account for victims that never enter a hospital. This places Ouellet and Kasantikul (2006), which attempts to reach all accident scenes rather than leveraging data on those admitted to hospitals, among the higher quality group of studies, although it still includes minimal corrections for selection into helmet usage (e.g. by safer drivers). In short, there are limitations to both Liu et al. (2008) and Ouellet and Kasantikul (2006) such that neither is clearly dominant, yet both are in line with the quality of evidence that practitioners have recently used in health campaigns.

Ouellet and Kasantikul (2006) leverages observational accident data. This study was selected because the authors also estimate that helmets are about 50% effective in Los Angeles, aligning well with the Liu et al. (2008) estimate from high-income settings. The specific estimate from

Ouellet and Kasantikul (2006) presented is based on helmets that stayed on. The authors note that when helmets were ejected, the mortality rate is higher. But, as presented in the study, this typically reflects incorrect helmet use and not effectiveness. Specifically, the study reports that about 6.8% of the unhelmeted riders died, versus 1.9% of the helmet-retained riders, resulting in the estimated efficacy of about 70% presented.

Respondents were presented with one study since interpreting two requires more time, and the variation helps to understand demand for safety. This should be thought of as two different draws from the distribution of how well helmets work, not one true and one false draw. The script for both studies informed respondents that estimates were from studies outside of Kenya so may not reflect Kenyan helmet effectiveness and that other studies may produce different results. The study received IRB approval from the University of California Committee on the Protection of Human Subjects and the Amref Health Africa IRB, which is based in Kenya.

An anonymous referee expressed the thoughtful concern that the intervention may have inadvertently harmed treated respondents by reducing their willingness to pay for helmets. I appreciate the referee's careful attention to the ethical dimensions of research. However, I interpret the findings differently. The observed decline in willingness to pay followed exposure to a well-regarded study on helmet effectiveness, which corrected overly optimistic priors. From this perspective, the intervention enabled more informed decision-making, leaving participants better off. In my view, withholding evidence on helmet effectiveness in order to encourage product uptake would raise its own ethical concerns. Moreover, the sample only included those not using helmets, so it did not cause anyone that was using a helmet to stop using one.

F Appendix Figures

A1: Survey locations

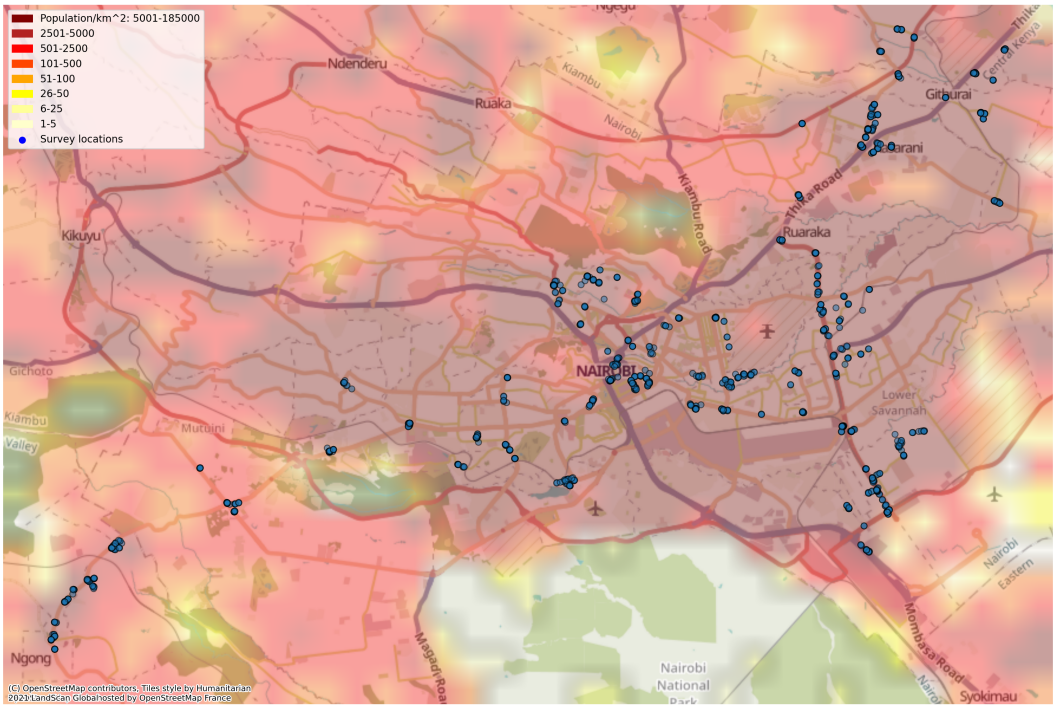
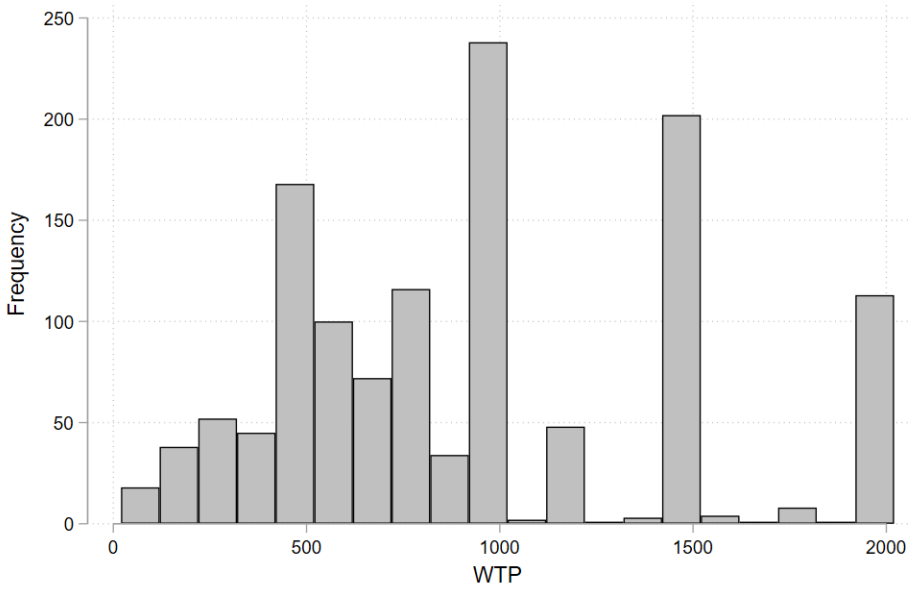


Figure A1 plots survey locations over a map of Nairobi. Color denotes population, where no color denotes no residents. The 2019 census estimated a population of about 4.4 million within the city. Map data is from Open Street Map. Population data is from the LandScan Global 2021 data set produced by Oak Ridge National Laboratory.

A2: Distribution of helmet bids (Kenyan shillings)

(a) A. Histogram of bids, restricted axis



(b) B. Histogram of bids, full distribution

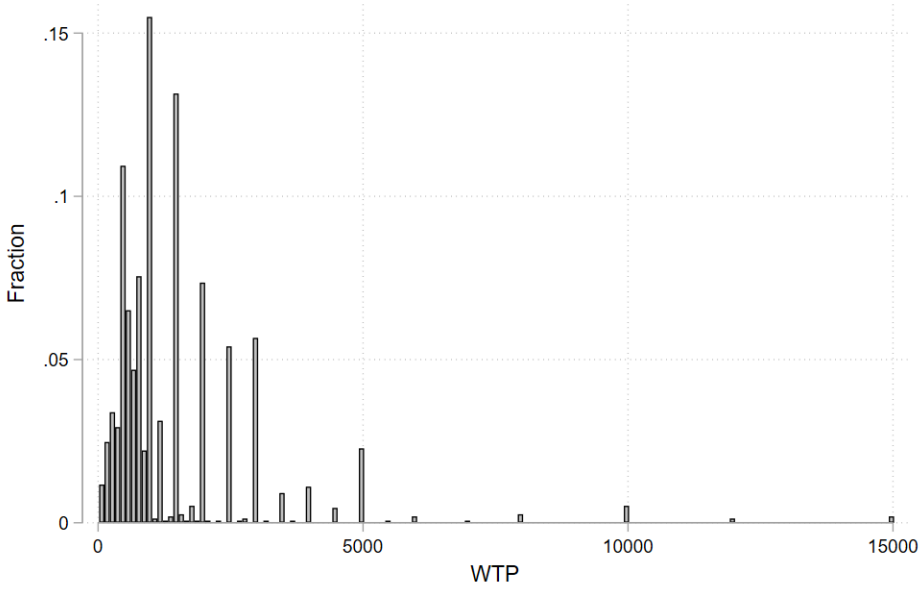


Figure A2 plots the distribution of willingness to pay for helmets in Kenyan shillings (Ksh). Panel A reports a histogram of bids, excluding outliers above Ksh 2,000 for clarity. Panel B reports the same histogram across the full distribution of bids. Figures present Ksh rather than USD to illustrate that respondents are more likely to select round numbers.

A3: Effect of study VSL on published benefit-cost ratios

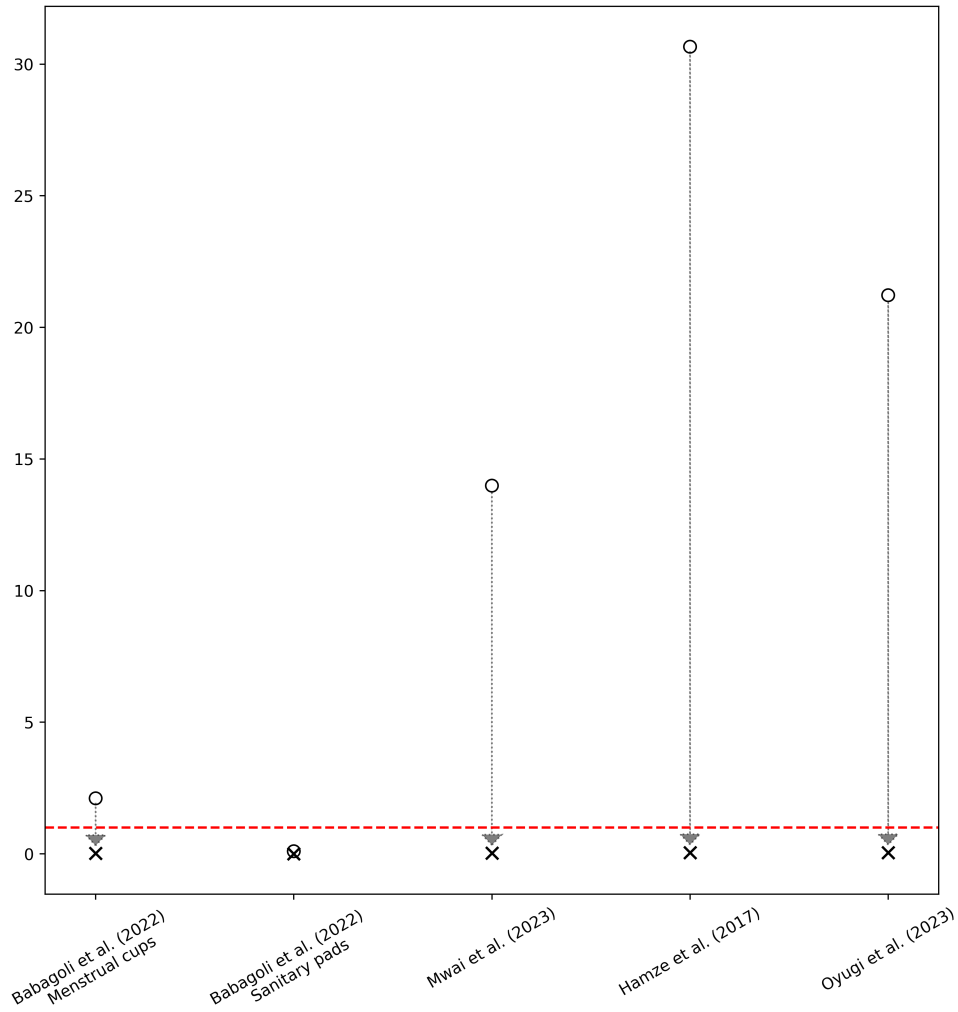
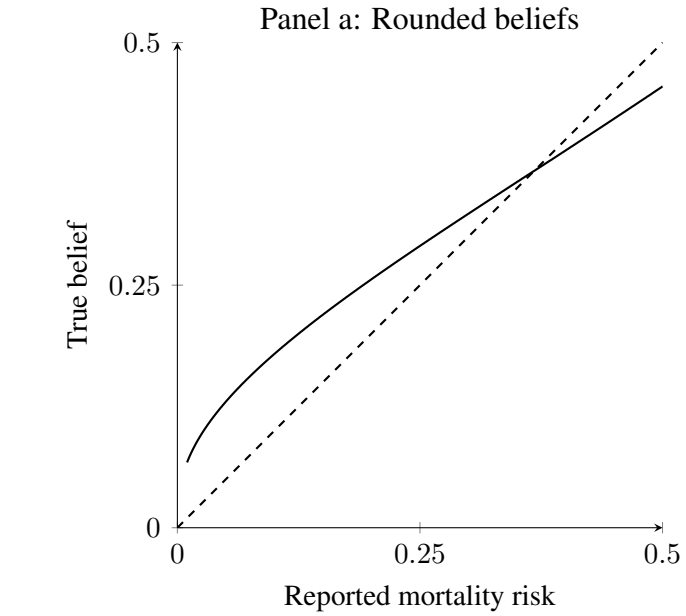
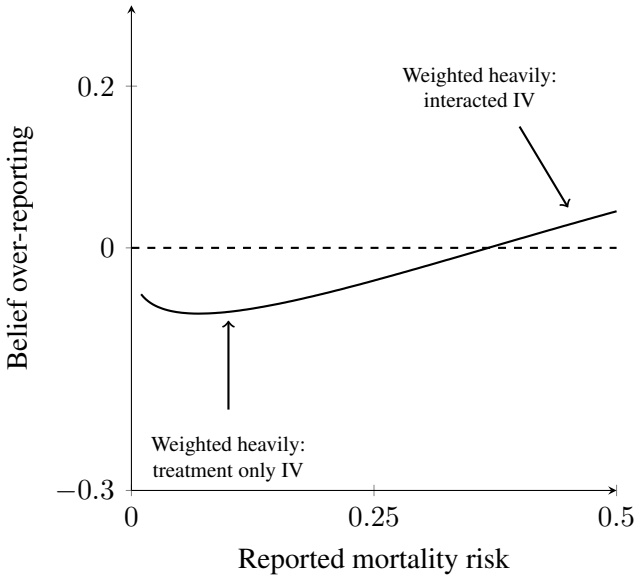


Figure A3 examines how the benefit-cost ratios (BCRs) of published benefit-cost analyses of Kenyan programs change when the study's original VSL or VSLY is replaced with the preferred estimates from this study. Hollow circles denote the original BCR estimate, and an x denotes the revised estimate. I estimate VSLY for use in this analysis using data on respondents' perceived life expectancy, which I use to derive the expected number of life years saved from a helmet. This allows for a revealed preference estimate of VSLY. The first two estimates are from Babagoli et al. (2022). The third estimate is from Mwai et al. (2023), the fourth is from Hamze et al. (2017), and the final figure is from Oyugi et al. (2023). The horizontal red line is at $BCR = 1$, the threshold for benefits exceeding costs.

A4: Belief distortion under rounding





Appendix Figure A4 illustrates how agents with low beliefs rounding down and those with high beliefs rounding up would produce different estimates between the "treatment only" and "interacted instruments." This figure plots a stylized case where agents that report they do not believe they face any risk of dying on a motorcycle act on a belief where they do face some risk, and those that report extremely high risks act on lower values. As shown in panel b, this would induce a positive correlation between misreporting and reported risk. The "interacted" instruments would therefore weight individuals with over-reported beliefs more heavily and underestimate VSL versus the "treatment only" instruments.

A5: Information sources used to form beliefs about motorcycle safety

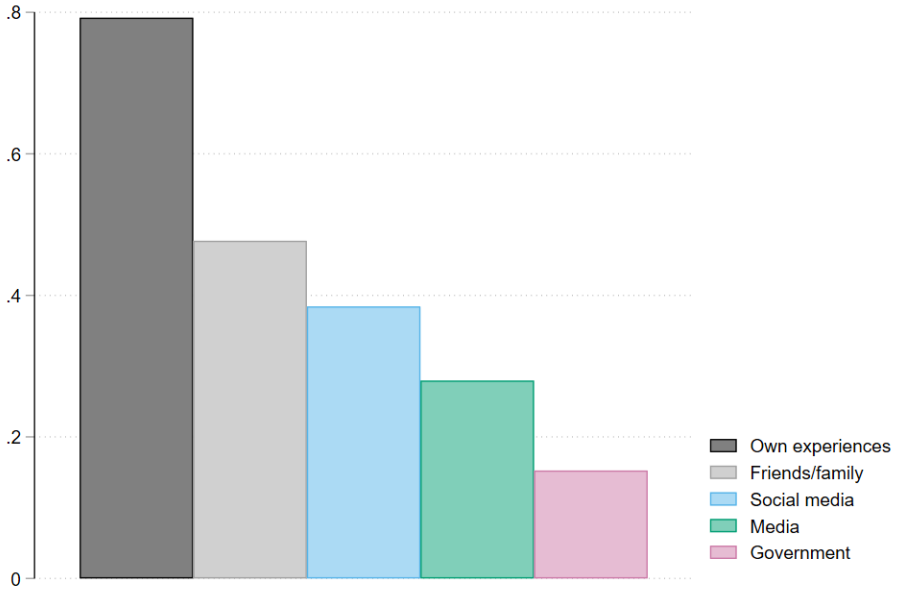
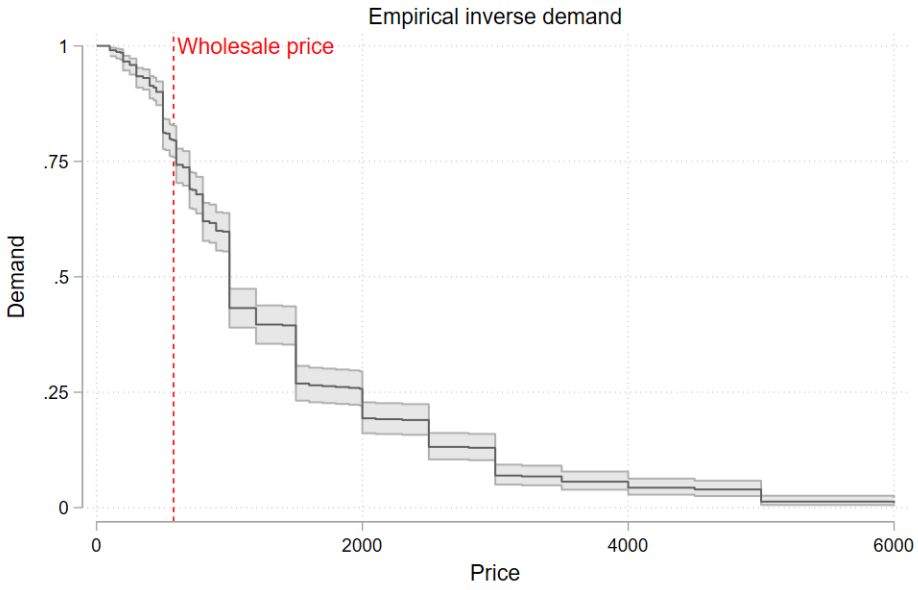


Figure A5 plots the information sources that respondents reported using to form beliefs about the mortality risk of motorcycles. Respondents were able to select multiple options, so the columns do not add to 1.

A6: Demand for helmets

(a) Inverse demand, control group



(b) Elasticity of demand, control group

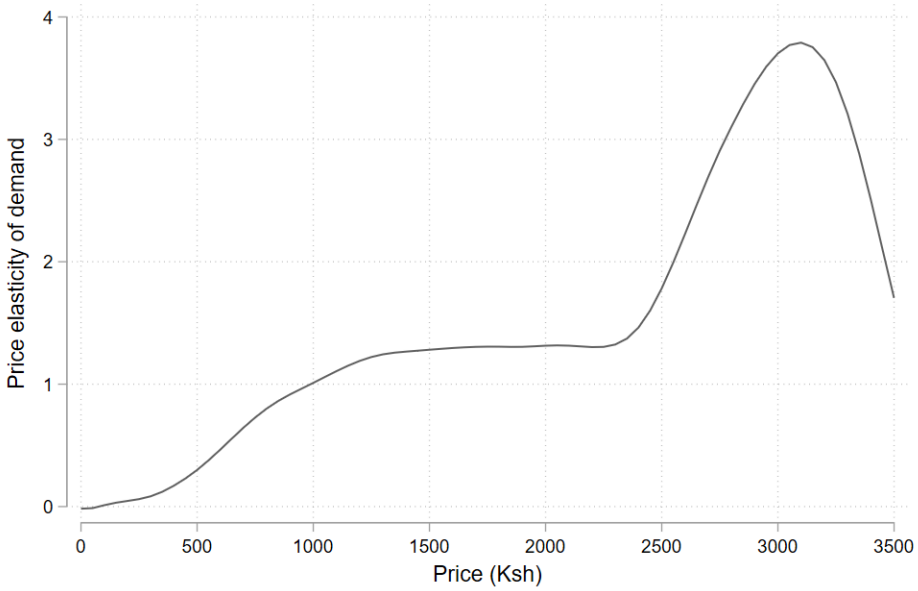


Figure A6 plots estimates of demand for helmets among observations in the control and pure control arms. The plot of demand includes a pointwise confidence interval. The plot of the elasticity of demand is based on a local polynomial estimation adapted from Berry et al. (2020). The vertical line denotes the wholesale price of helmets, which was Ksh 580 during the study. The figures exclude 35 observations for which surveyors reported motorcycle taxi drivers pretending to be passengers.

A7: Inverse demand, pure control and control group

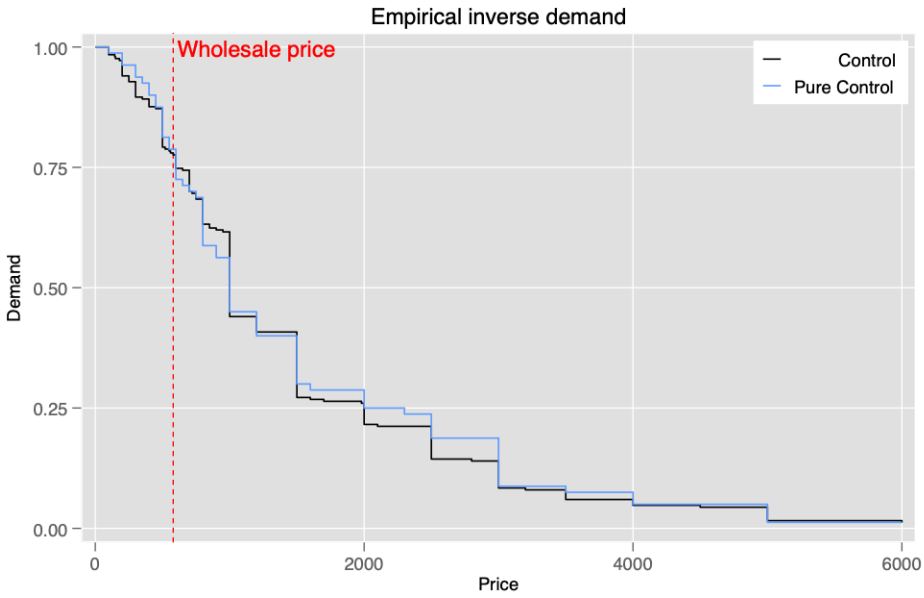


Figure A7 plots estimates of demand for helmets among observations in the control and pure control arms. The vertical line denotes the wholesale price of helmets, which was Ksh 580 during the study. The figures exclude 35 observations for which surveyors reported motorcycle taxi drivers pretending to be passengers. Observations from wave 2 are excluded, since the pure control was excluded from that wave.

G Appendix Tables

A1: Summary statistics and balance: Motorcycle taxi use

	(1) Control Summary statistics	(2) Pure control - control	(3) Low treatment - Control	(4) High treatment - Control	(5) High - low treatment
Trips/week	7.531	1.178	0.196	-0.379	-0.593*
	[5.242]	(0.725)	(0.348)	(0.358)	(0.343)
Average trip length (minutes)	19.593	0.334	-0.704	-1.301*	-0.583
	[12.567]	(1.538)	(0.725)	(0.745)	(0.672)
$\mathbb{E}$[Deaths/10,000 passengers, 1 year]	371.584	NA	-46.930	-0.326	46.539
	[1,186.706]	NA	(76.351)	(78.569)	(76.478)
$\mathbb{E}$[Deaths/10,000 passengers, 5 years]	977.118	NA	-154.821	266.926	420.383**
	[2,479.942]	NA	(192.351)	(197.928)	(205.063)
Confidence in beliefs	3.461	NA	0.034	0.011	-0.022
	[0.699]	NA	(0.042)	(0.044)	(0.041)
10000*Pr(Fatal accident, 5 years)	354.133	NA	-19.377	-40.568	-20.464
	[973.710]	NA	(56.011)	(57.630)	(52.741)
Previous accident	0.488	NA	-0.011	-0.030	-0.019
	[0.500]	NA	(0.032)	(0.033)	(0.032)
Know accident victim	0.947	NA	-0.022	-0.036**	-0.014
	[0.224]	NA	(0.017)	(0.017)	(0.017)
Use motorcycle taxi: Commuting	0.781	0.027	-0.032	0.008	0.038
	[0.414]	(0.054)	(0.027)	(0.028)	(0.027)
Shopping	0.420	-0.031	-0.033	-0.018	0.016
	[0.494]	(0.064)	(0.031)	(0.032)	(0.031)
Leisure	0.261	0.057	0.074**	0.036	-0.037
	[0.440]	(0.058)	(0.029)	(0.030)	(0.029)
Deliveries	0.095	-0.012	0.015	-0.000	-0.015
	[0.294]	(0.014)	(0.018)	(0.018)	(0.018)
Emergency/hospital transportation	0.095	-0.008	-0.009	-0.007	0.003
	[0.294]	(0.014)	(0.017)	(0.017)	(0.017)
Reason for use: Speed	0.816	0.114**	0.019	0.003	-0.017
	[0.388]	(0.045)	(0.024)	(0.025)	(0.024)
Convenience	0.717	-0.005	0.005	-0.045	-0.049*
	[0.451]	(0.061)	(0.029)	(0.030)	(0.029)
Only option	0.168	0.007	-0.025	-0.010	0.015
	[0.374]	(0.047)	(0.023)	(0.024)	(0.023)
Price	0.106	0.037	-0.008	-0.026	-0.018
	[0.308]	(0.038)	(0.019)	(0.019)	(0.018)
Safety/Avoid dangerous areas	0.069	-0.017	-0.027*	-0.010	0.017
	[0.253]	(0.033)	(0.015)	(0.015)	(0.014)
Enjoyment	0.011	-0.012	-0.002	0.008	0.009
	[0.105]	(0.015)	(0.007)	(0.007)	(0.007)
Risk information: Own experiences	0.819	NA	-0.036	-0.042	-0.007
	[0.386]	NA	(0.026)	(0.027)	(0.026)
Friends/family	0.454	NA	0.029	0.038	0.009
	[0.498]	NA	(0.032)	(0.033)	(0.032)
Social media	0.414	NA	-0.049	-0.034	0.016
	[0.493]	NA	(0.031)	(0.032)	(0.031)
Media	0.288	NA	0.014	-0.027	-0.042
	[0.453]	NA	(0.029)	(0.029)	(0.028)
Government	0.135	NA	0.015	0.026	0.011
	[0.342]	NA	(0.023)	(0.023)	(0.023)
N (exc. control in col. 2-4)	452	80	531	473	
p-value of joint orthogonality		0.008	0.136	0.221	0.221

Standard deviations in brackets. Standard errors in parentheses.

Column (1) reports the mean and standard deviation of the indicated variable across the control group. Columns (2) - (5) report the difference in mean of each variable across treatment arms, where terms in parentheses are standard errors of this pairwise t-test. Missing values are imputed using the median across the full sample for the joint test.

A2: Elasticity of helmet demand with respect to subjective beliefs

	All observations		Information control	
	(1)	(2)	(3)	(4)
	Motorcycle risk	Motorcycle risk	Helmet effectiveness	Helmet effectiveness
Elasticity	0.012	0.014	0.420	0.441
	(0.007)	(0.007)	(0.161)	(0.159)
Controls	No	Yes	No	Yes
Observations	1,421	1,421	452	452

Robust Standard errors in parentheses.

Table A2 reports the elasticity of willingness to pay for a helmet with respect to priors about the likelihood of suffering a fatal accident in columns (1) - (2) and perceived efficacy of a helmet at preventing death among the information control arm in columns (3) - (4). Columns (2) and (4) include controls for the log of income, digit span recall, and log years of education. Estimates exclude 35 observations for which surveyors reported motorcycle taxi drivers pretending to be passengers.

A3: Value of a statistical life: Winsorized beliefs

	Raw WTP				Winsorized WTP	
	(1) Interacted	(2) Treatment only	(3) Interacted	(4) Treatment only	(5) Interacted	(6) Treatment only
VSL	1,022.04	671.97	1,054.78	653.89	712.40	436.77
	(504.14)	(391.26)	(522.87)	(426.06)	(374.03)	(324.33)
Kleibergen-Paap F-stat	4.89	8.98	4.92	7.78	4.92	7.78
Weak IV Robust Confidence Set	[163.69, 2,399.38]	[-56.16, 1,833.88]	[164.55, 2,483.29]	[-172.73, 2,054.10]	[75.57, 1,526.95]	[-166.81, 1,374.25]
Inversion test	CLR	CLR	CLR	CLR	CLR	CLR
Observations	1,425	1,425	1,425	1,425	1,425	1,425
Controls	BL Risk	BL Risk	LASSO	LASSO	LASSO	LASSO
Enumerator FE	Yes	Yes	Yes	Yes	Yes	Yes

Robust standard errors in parentheses.

Columns labeled "Interacted" use a first stage consisting of treatment indicators and treatment indicators interacted with baseline motorcycle risk beliefs as instruments for the mortality risk reduction of a helmet. Columns labeled "Treatment only" use only treatment assignment as an instrument. All models control for baseline risk beliefs. Columns (5) - (6) report estimates where willingness to pay is winsorized at the 2nd and 98th percentiles. Over-identified models report weak instrument robust confidence sets constructed using inversion of a conditional likelihood ratio (Moreira, 2003). For just-identified models, weak instrument robust confidence sets are constructed by inverting an AR test (Anderson and Rubin, 1949). All models exclude 35 observations for which surveyors reported motorcycle taxi drivers pretending to be passengers. Beliefs about the mortality reduction from a helmet are winsorized at the 2nd and 98th percentiles.

A4: Robustness of VSL to alternative assumptions

Panel A: Change in planned future ridership

	Raw WTP				Winsorized WTP	
	(1) Interacted	(2) Treatment only	(3) Interacted	(4) Treatment only	(5) Interacted	(6) Treatment only
VSL	307.83	282.37	326.85	264.27	223.82	199.55
	(132.52)	(188.60)	(130.83)	(195.49)	(96.08)	(144.70)
Cragg-Donald F-stat	10.90	10.21	11.21	9.52	11.21	9.52
Weak IV Robust Confidence Set	[75.40, 716.79]	[-91.51, 801.91]	[102.42, 730.28]	[-130.83, 811.08]	[54.46, 502.22]	[-86.79, 578.53]
Inversion test	CLR	CLR	CLR	CLR	CLR	CLR
Observations	1,425	1,425	1,425	1,425	1,425	1,425
Controls	BL Risk	BL Risk	LASSO	LASSO	LASSO	LASSO

Panel B: Different beliefs about helmet lifespan

	Raw WTP				Winsorized WTP	
	(1) Interacted	(2) Treatment only	(3) Interacted	(4) Treatment only	(5) Interacted	(6) Treatment only
VSL	469.60	332.13	491.66	298.61	347.25	240.87
	(162.99)	(245.11)	(162.87)	(246.66)	(118.41)	(182.05)
Cragg-Donald F-stat	13.03	10.38	13.06	10.06	13.07	10.06
Weak IV Robust Confidence Set	[195.81, 939.80]	[-151.15, 1,032.40]	[220.04, 960.00]	[-198.32, 988.80]	[141.71, 666.81]	[-118.93, 712.36]
Inversion test	CLR	CLR	CLR	CLR	CLR	CLR
Observations	1,411	1,425	1,411	1,425	1,411	1,425
Controls	BL Risk	BL Risk	LASSO	LASSO	LASSO	LASSO

Panel C: Weighted by 1/rides per week

	Raw WTP				Winsorized WTP	
	(1) Interacted	(2) Treatment only	(3) Interacted	(4) Treatment only	(5) Interacted	(6) Treatment only
VSL	361.00	96.56	366.01	86.40	234.72	110.97
	(146.50)	(311.00)	(146.47)	(304.39)	(107.54)	(225.65)
Cragg-Donald F-stat	20.12	7.97	19.81	8.26	19.81	8.26
Weak IV Robust Confidence Set	[82.68, 691.70]	[-637.11, 865.77]	[88.88, 699.07]	[-632.16, 837.48]	[27.21, 470.51]	[-388.60, 671.41]
Inversion test	CLR	CLR	CLR	CLR	CLR	CLR
Observations	1,425	1,425	1,425	1,425	1,425	1,425
Controls	BL Risk	BL Risk	LASSO	LASSO	LASSO	LASSO

Standard errors in parentheses.

Panel A accounts for increases in planned future ridership associated with receiving a helmet. For those that received a cash gift, this value is imputed by regressing planned future ridership on past ridership interacted with treatment assignment. Panel B uses respondents' stated beliefs about the lifespan of the helmet, rather than the manufacturer's suggestion. Panel C weights each observation by the inverse of motorcycle trips/week to account for selection into ridership. All models control for baseline beliefs and enumerator fixed effects. Excludes 35 observations where motorcycle taxi drivers pretended to be passengers.

A5: Value of a statistical life: GMM and IV probit estimates with robust standard errors

	GMM: Raw WTP		GMM: Winsorized WTP		IV probit	
	(1) Interacted	(2) Treatment only	(3) Interacted	(4) Treatment only	(5) Treatment only	(6) Treatment only
VSL	464.03	308.12	353.47	252.59	161.25	161.92
	(228.88)	(238.51)	(178.48)	(193.40)	(118.35)	(118.42)
Price	NA	NA	NA	NA	-0.13	-0.13
					(0.03)	(0.03)
Risk reduction	NA	NA	NA	NA	20.31	20.71
					(10.17)	(10.40)
Kleibergen-Paap F-stat	6.18	10.96	6.18	10.96	NA	NA
Weak IV Robust Confidence Set	[1.85, 1,306.84]	[-154.63, 978.63]	[7.19, 883.49]	[-122.64, 750.35]	NA	NA
Inversion test	CLR	CLR	CLR	CLR	NA	NA
Observations	1,425	1,425	1,425	1,425	1,425	1,425
Controls	LASSO	LASSO	LASSO	LASSO	BL Risk	LASSO
Enumerator FE	Yes	Yes	Yes	Yes	Yes	Yes

Robust Standard errors in parentheses and weak instrument robust confidence sets in brackets.

Table A5 reports VSL estimates under heteroskedastic robust standard errors using continuous updating GMM (CUE) in columns (1)-(4) and IV probit, estimated via maximum likelihood, in columns (5)-(6). Weak instrument robust confidence sets are calculated using conditional likelihood ratio test inversion. Columns labeled "Interacted" use a first stage consisting of treatment indicators and treatment indicators interacted with baseline motorcycle risk beliefs as instruments for the mortality risk reduction of a helmet. Columns labeled "Treatment only" use only treatment assignment as an instrument. All models control for baseline risk beliefs. Columns (3) - (4) report estimates where willingness to pay is winsorized at the 2nd and 98th percentiles. IV probit estimates are estimated by fitting a regression of an indicator for receiving a helmet on the randomly drawn price (from the BDM game) and risk reduction, instrumented for with treatment assignment. Interacted instruments are not used for these estimates since the effective number of observations is smaller and the arguments supporting that instrument depend on linearity. Estimates exclude 35 observations for which surveyors reported motorcycle taxi drivers pretending to be passengers.

A6: Correlations between perceived mortality risk and demographics

	(1) Years of education	(2) Female	(3) log(Income)	(4) Age	(5) Above-median health	(6) 1(Children)
Coefficient	2.20	-95.56	43.83	-1.20	-57.59	53.09
	(8.15)	(49.46)	(29.20)	(2.37)	(46.69)	(52.22)
Sample average	333.22	333.22	333.22	333.22	333.22	333.22
Observations	1,419	1,427	1,221	1,425	1,427	1,427

Robust Standard errors in parentheses.

Table A6 reports coefficients of regressions of perceived baseline mortality risk on demographic covariates. As detailed in the results, the local average VSL identified by this study weights observations with higher perceived risk more heavily, so this table aims to understand whether the LATE is over a selected population. Column (1) considers years of education, (2) an indicator equal to 1 if the respondent is female, (3) the log of income, (4) respondent age, (5) an indicator for reporting above-median health, and (6) an indicator for having children. No controls are included since the aim is only to understand what covariates are correlated with perceived risk. Estimates exclude 35 observations for which surveyors reported motorcycle taxi drivers pretending to be passengers.

A7: Effect of low treatment on WTP

	(1) Full sample	(2) Excluding Pure control	(3) Prior: $< .01$	(4) Prior: $\geq .01$
Low treatment	-3.41	-3.41	-2.04	-5.83
	(1.59)	(1.62)	(1.99)	(2.95)
Observations	1,536	1,456	1,064	363
Controls	LASSO	LASSO	LASSO	LASSO
Pure control included	Yes	No	No	No

Robust Standard errors in parentheses.

This table reports effects of assignment to the low treatment arm on helmet willingness to pay. Other arms are pooled since beliefs are similar. Estimates exclude 35 observations for which surveyors reported motorcycle taxi drivers pretending to be passengers. Column (1) includes pure control observations, (2)-(4) exclude them. All estimates include enumerator fixed effects and covariates selected via single-post LASSO. Column (3) includes observations with priors about the risk of dying in a motorcycle accident below .01, column (4) includes only observations where priors are at least .01.

A8: VSL Estimates: Robustness to sample and extreme beliefs

	(1) PAP controls	(2) Employed only	(3) Imputed Risk	(4) Top coded: Treatment only	(5) Top coded: Interacted	(6) Driver surveys included
VSL	593.35	678.17	144.79	654.10	1,032.32	512.96
	(196.02)	(210.39)	(79.28)	(490.92)	(449.44)	(200.03)
Cragg-Donald F-stat	11.16	10.75	14.73	6.70	4.83	11.00
Weak IV Robust	[255.72,	[343.84,	[-7.40,	[-271.97,	[372.87,	[169.47,
Confidence Set	1,130.90]	1,323.49]	333.77]	2,364.92]	3,150.73]	1,087.04]
Inversion test	CLR	CLR	CLR	CLR	CLR	CLR
Instruments	Int	Int	Int	TO	Int	Int
Observations	1,425	1,237	1,427	1,425	1,425	1,455
Controls	LASSO	LASSO	No	No	No	LASSO

Standard errors in parentheses.

Table A8 reports VSL estimates under covariate, sample or belief restrictions for robustness. Column (1) reports the interacted instrument using the covariate set defined in the original PAP. Column (2) reports interacted estimates restricting the sample to the employed. Column (3) reported interacted estimates in which I estimate beliefs about the probability a helmet saves one's life from beliefs about deaths per 10,000 passengers over 5 years multiplied by helmet effectiveness, allowing integer rather than probability responses. I use the actual probability response in the interacted instrument: point estimates are similar but standard errors are larger with the treatment only IV. Columns (4) - (5) top code priors and beliefs about the likelihood a helmet saves one's life at the 90th percentile of these variables distributions. Column (6) includes observations where motorcycle drivers pretended to be passengers and submitted false surveys, which are excluded from other estimates. "Int" denotes the interacted instrument set and "TO" treatment only.